

Technical University of Munich

[SCHOOL OR DEPARTMENT]

Audited Semantic Propagation in LLM-Agent Societies

Persona Composition, Network Topology, and Climate Firestorms

Master's Thesis

Candidate:	[CANDIDATE NAME]
Student ID:	[MATRICULATION NUMBER]
Programme:	[DEGREE PROGRAMME]
First examiner:	[FIRST EXAMINER]
Supervisor:	[SUPERVISOR NAME]
Advisor:	[ADVISOR NAME, IF APPLICABLE]
Submission date:	[SUBMISSION DATE]

Abstract

Online firestorms can connect scientific proposals with established misinformation narratives. This thesis examines how factual content changes when large language model (LLM) agents transmit climate and geoengineering messages. It extends a prior auditor–node instrument with persona compositions, eight generated topologies, and a separate 63-node complex network. Four research questions address measurement, persona composition, topology, and empirical anchoring.

The canonical Phase 2 campaign used `gpt-4o-mini`, eight hops, one run per configuration, 288 configurations, and 1,728 article cells. It retained 292 cells with at most one scored event as hatched dead cascades. Continuous and dual-discrete Misinformation Propagation Rate (MPR) remained separate. Dual-discrete scores exceeded continuous scores on every topology aggregate, which demonstrates instrument sensitivity rather than a common-scale increase. Conspiracy-oriented personas and conspiracy-heavy mixes scored above scientific personas and the conspiracy-free mix in both instruments. Topology conditioned these contrasts, but no universal ranking emerged.

The D-net experiment applied the same protocol to a 63-node hashtag co-occurrence graph. Conspiracy-only assignments exceeded mixed assignments on both articles and instruments. Structural validation failed overall. The graph was not a hydrated retweet cascade, empirical timing was unavailable, and Debnath et al. provided no empirical Twitter MPR. The results therefore support a reproducible mechanism probe, not a causal account of human behaviour or a validated digital twin.

Individual contribution and prior-work boundary

The auditor–node engine, MI, MPR, and the general-news linear-chain study are co-authored prior work (Maurya et al., 2025). The climate corpus, Phase 2 campaign, Pfeffer operationalisation, composition analysis, and D-net boundary test form the thesis research layer.

Unresolved placeholder: the candidate must insert the exact sentence-level division of code, experimental design, analysis, and writing among all co-authors before submission.

Declaration

Official wording required. Insert the current statutory declaration from the responsible TUM examination office here. The official text, place, date, signature fields, and any generative-AI disclosure were not available in the source material and have not been invented.

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Abbreviations

D-net	Separate 63-node custom complex network.
DTFS	Digital Twin Fidelity Score.
H	Homogeneous persona assignment.
He	Heterogeneous persona assignment.
IFD	Information Fidelity and Distortion record.
LLM	Large language model.
MI	Misinformation Index.
MPR	Misinformation Propagation Rate.
T2c	Phase 2 continuous-instrument campaign.
T2d	Phase 2 dual campaign with a discrete headline.
k^*	First tick above MI 3 with no later observed recovery.

1 Introduction

1.1 Problem and motivation

Online misinformation changes as people repeat, reinterpret, and contest it. This process matters when a scientific proposal becomes attached to an established conspiracy narrative. Solar geoengineering provides such a case. Public discussion of stratospheric aerosol intervention and Harvard’s SCoPEX project has intersected with claims about chemtrails, weather control, and elite manipulation (Debnath et al., 2023b). These claims can affect how citizens interpret research, institutions, and climate policy. The relevant problem is therefore not only whether false content travels. It is also how network structure and social identity shape factual change during transmission.

Online firestorm theory offers a social account of rapid, affective, and clustered communication. Pfeffer, Zorbach, and Carley describe an online firestorm as a sudden volume of negative word of mouth directed at a person, company, or group (Pfeffer, Zorbach, and Carley, 2014). Their Outlook identifies seven interrelated factors: speed and volume, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. The theory explains why particular communication environments can intensify collective reactions. It does not specify a generative agent model or a quantitative tipping rule.

Geoengineering discourse supplies an empirical setting in which several of these mechanisms may matter. Debnath and colleagues analysed 814,924 English-language tweets containing the geoengineering hashtag between 2009 and 2021 (Debnath et al., 2023b). They identified a prominent chemtrails discourse alongside climate-action and environmental-concern themes. They also documented temporal changes around solar-radiation-management projects and governance events. Their study maps observed discourse at scale. It does not provide controlled counterfactuals in which the same message enters different networks or identity compositions.

Agent simulation can provide such control, but it introduces a separate validity problem. Classical cascade models represent diffusion through states or activation probabilities (Kempe, Kleinberg, and Tardos, 2003). They can isolate structural mechanisms, but they do not directly represent claim-level changes in natural language. Large language model agents can transform text while retaining persona and interaction context. Their outputs, however, reflect prompts, model behaviour, and scoring choices rather than observed human conduct. Generative social simulations must therefore expose their assumptions and restrict claims to the evidence that their design supports (Sargent, 2013).

Maurya and colleagues provide the immediate technical basis for this thesis. Their LASS workshop paper at ACM CIKM 2025 introduced an auditor-node framework for persona-conditioned rewrites on linear chains (Maurya et al., 2025). The framework scores factual

degradation through a misinformation index (MI) and aggregates scores through a misinformation propagation rate (MPR). That study used general-news domains, fixed chains of up to 30 rewrites, and a discrete question-answering auditor. Its framework and published findings are prior work. This thesis neither reintroduces those metrics nor reuses its crime and commercial results as new evidence.

1.2 Research gap

The three preceding research strands leave a specific methodological gap. Pfeffer et al. provide a theory of firestorm conditions without a generative testbed. Debnath et al. provide an observational account of geoengineering discourse without controlled interventions on network structure or identity composition. Maurya et al. provide a generative misinformation instrument without a climate corpus, a graph-topology comparison, or an explicit mapping to firestorm theory. The gap addressed here lies at their intersection. It concerns how to design and audit a controlled climate-firestorm simulation without treating simulation output as observed social behaviour.

The proposal framed this intersection as six factors: valence, surprise, identity alignment, network clustering, information echo, and temporal acceleration. Those labels are not Pfeffer et al.'s original seven-factor list. This thesis treats them as a computational remapping. It also records the parts that the simulation does not implement. Cross-media dynamics has no media-broadcast layer. Binary platform choices have no direct analogue because agents may forward, reinterpret, or drop a message. Surprise remains a single-node drip seed, and the temporal horizon remains fixed. The study therefore evaluates selected mechanisms rather than claiming to reproduce the complete firestorm theory.

Measurement creates a second gap inside the simulation. The prior discrete auditor counts omitted and contradicted answers. The present campaign also uses a continuous factual-recovery score. These instruments share an approximate range but not a common measurement scale. A difference between their outputs may reflect scoring design rather than a different social process. The analysis must therefore keep discrete and continuous MI separate.

Empirical grounding creates a third constraint. The public Debnath artefacts available to this study did not yield hydrated tweet text or conversation cascades. The thesis consequently uses theory-faithful behavioural-profile reductions. It also uses a 63-node hashtag co-occurrence graph as a structural fallback. Neither object is an empirical user cluster or a retweet network. This distinction prevents the simulation from being described as a digital twin of Twitter.

1.3 Aim and analytical approach

This thesis aims to develop and evaluate an exploratory artificial-society testbed for climate-misinformation propagation. The testbed extends the prior auditor-node framework with climate and geoengineering seeds, Debnath-grounded behavioural profiles, and configurable network topologies. It maps the firestorm theory to simulation controls, held conditions, and measured outcomes. It then tests how identity composition, topology, and auditor design

shape measured factual degradation.

The main campaign crosses eight graph topologies with two identity designs and two scoring modes. The topologies are a linear chain, ring, Erdős–Rényi graph, small-world graph, scale-free graph, echo chamber, polarised graph, and hierarchy. The homogeneous arm assigns one of 12 behavioural profiles to all eight nodes. The heterogeneous arm assigns one of six eight-profile mixtures. Each configuration processes six climate or geoengineering articles over eight ticks. The resulting design contains 1,728 article-level cells. All cells use `gpt-4o-mini`, one graph seed, and one stochastic run.

The analysis reports MI and mean node MPR within each scoring mode. It defines k^* as the first tick at which mean MI exceeds 3 and remains above 3 for every later observed tick. This quantity is a right-censored simulator clock. It is not an estimate of ignition time on social media. The analysis also retains 292 cells with at most one scored event as hatched dead cascades. It excludes those cells from live-cell means but does not recode them as zero.

A separate D-net analysis places the same agents on the 63-node hashtag graph. This analysis compares simulated cascade shape with the documented fallback structure. It also compares homogeneous conspiracy profiles with a mixed profile assignment on the same edge set. The comparison concerns graph structure and simulated auditor scores. Debnath et al. report no empirical MI or MPR against which those scores could be validated.

1.4 Research questions

The study addresses four research questions. Each question follows from a design choice that the available evidence can evaluate.

RQ1 Holding identity arm and topology labels fixed, how do the continuous and dual-discrete auditor instruments change estimated misinformation severity, k^* occurrence, and topology ordering?

RQ1 treats scoring mode as a measurement question. It compares modes without averaging their values or interpreting their gap as a third misinformation measure.

RQ2 Within one scoring mode, does explicit conspiracy composition explain misinformation severity and k^* occurrence more consistently than the coarse homogeneous or heterogeneous label?

RQ2 compares persona families and named mixtures. It treats composition as substantive and H or He as an assignment description.

RQ3 How does topology condition the same-mode relationship between persona composition, live-cell MI, and irreversible k^* within the eight-tick horizon?

RQ3 compares the eight generated topologies. It reports dead-cell rates with live-cell means because cascade survival changes the observed population.

RQ4 What can the 63-node hashtag-based D-net comparison establish about structural similarity and identity-conditioned factual degradation, and where does that comparison fail?

RQ4 separates empirical graph properties from simulated content scores. It tests a documented fallback rather than claiming reconstruction of Debnath et al.’s tweet cascades.

These questions support descriptive comparisons. They do not support population estimates or causal effects. The single stochastic run also precludes confirmatory inference about topology, identity, or scoring mode.

1.5 Contributions

This thesis makes four bounded contributions.

1. It provides an explicit mapping from Pfeffer et al.’s seven firestorm factors to the simulation’s varied, held, measured, and absent elements. The mapping corrects the proposal’s six-factor shorthand and retains cross-media dynamics as an unimplemented boundary.
2. It constructs a documented climate and geoengineering simulation layer for the prior auditor–node framework. This layer contains six campaign articles, 12 homogeneous behavioural profiles, six heterogeneous mixtures, eight topologies, and reproducible configuration files. The profiles reduce themes reported by Debnath et al.; they are not clusters estimated from 814,924 tweets.
3. It evaluates identity composition and network topology under two distinct factual-degradation instruments. The analysis preserves dead cascades, separates continuous from discrete scores, and distinguishes persona composition from the broad homogeneous–heterogeneous contrast.
4. It links a 63-node hashtag co-occurrence structure to a custom simulation graph and audits the limits of that comparison. This analysis reports structural mismatch as well as similarity. It does not equate auditor MI with Twitter behaviour.

These contributions concern operationalisation, experimental design, and transparent measurement. They do not include the original auditor–node architecture, MI, MPR, or the published linear-chain findings. They also do not establish a validated model of public response to geoengineering.

1.6 Scope and limitations

The empirical scope is deliberately narrow. The campaign uses six authored or curated climate articles and a finite set of prompt-based behavioural profiles. It runs eight-node graphs for eight ticks with one model and one graph seed. The design changes text through LLM role-play. It does not model stable human belief, individual psychology, or platform recommendation systems.

The data scope is narrower than the proposal anticipated. The study did not hydrate the 814,924 tweet identifiers. It did not estimate HDBSCAN profiles, reconstruct conversation trees, or reproduce Debnath et al.’s temporal series. The hashtag graph records co-occurrence structure, not retweet transmission. The simulation clock cannot be converted into minutes, days, or empirical hop distance.

The inferential scope follows these design limits. The results describe one stochastic campaign. They do not warrant significance claims, robust rankings, or general laws about homogeneous and heterogeneous groups. The two auditors remain model-based measurements. Their disagreement is evidence about the instrument, not an independent factual ground truth. The k^* measure is conditional on the eight-tick observation window. A value of “none” means that no irreversible crossing appeared within that window.

1.7 Thesis roadmap

Chapter 2 develops the theoretical and empirical background. It covers online firestorms, climate and geoengineering discourse, diffusion models, LLM-agent simulation, and automated factuality assessment. Chapter 3 maps Pfeffer et al.’s seven factors to the thesis design. Chapter 4 separates the inherited system from the climate-specific design and documents the articles, behavioural profiles, and hashtag graph. Chapter 4 specifies the factorial campaign, both auditor modes, dead-cell rule, and k^* . Chapter 5 answers RQ2 and RQ3 across the 1,728-cell campaign. It then answers RQ4 through the D-net comparison. Chapter 6 interprets the findings against Pfeffer, Debnath, and the prior linear-chain study. Chapter 7 addresses validity, ethics, and reproducibility. Chapter 8 answers the research questions and states the conditions required for stronger claims.

The thesis therefore proceeds from theory to operationalisation, from operationalisation to measurement, and from measurement to bounded interpretation. Its central claim is methodological. An LLM society can expose how climate-misinformation results depend on identity, topology, and the scoring instrument, but this campaign cannot stand in for the public it simulates.

2 Literature Review

This chapter locates the thesis at the intersection of misinformation diffusion, online-firestorm theory, geoengineering discourse, network science, and large-language-model (LLM) agent simulation. The distinction between these literatures matters. A model may reproduce the reach of a cascade without representing how its claims change, while fluent agent output may look socially plausible without constituting a valid model of people. The review therefore connects substantive theory to the thesis’s observables and closes by stating the specific gap addressed by the experiments.

2.1 Misinformation as diffusion and semantic transformation

Misinformation is not a single technical object. The term can refer to false content, an actor’s belief, an exposure, a sharing decision, or a population-level diffusion pattern. Lazer et al. (2018) consequently call for research that joins individual cognition, institutions, and platform structure rather than treating falsity as a classification problem alone. Large-scale observational evidence confirms that diffusion is socially structured: Vosoughi, Roy, and Aral (2018) found that false news on Twitter travelled farther, faster, deeper, and more broadly than true news in their data. Novelty and emotional response helped distinguish false from true cascades, while automated accounts accelerated both. These findings motivate models with content and agent state, but they do not imply that every simulated cascade should reproduce the same ordering.

Classical diffusion models provide useful null formalisms. In the independent cascade model, an activated node receives one probabilistic opportunity to activate each neighbour (Kempe, Kleinberg, and Tardos, 2003). Related compartmental and threshold models reduce the state of a node to susceptible, active, adopted, or an opinion scalar. Their abstraction is an advantage for identifying conditions for reach or consensus, but an edge transmits a state rather than rewriting a proposition. This thesis instead studies a transmission in which the text itself can lose, invert, or acquire claims. The distinction is between *whether* something propagates and *what it becomes while propagating*.

Empirical cascade size is also not synonymous with virality. Goel et al. (2016) show that similarly popular diffusion events range from broadcast-like stars to multigenerational chains and define structural virality from pairwise distances in a diffusion tree. A scale-free contagion model did not recover the full empirical diversity they observed. This is a warning against explaining an outcome from a topology label alone. In the present study, event count, semantic degradation, and time to a severity threshold are therefore conceptually separate outcomes.

2.2 Online firestorms: theory and operationalisation

Pfeffer, Zorbach, and Carley (2014) define an online firestorm as a sudden discharge of large quantities of negative word-of-mouth and complaint behaviour directed at a person, organisation, or group. Their contribution is a qualitative synthesis of the mechanisms that make rapid, affective outrage possible. The paper’s outlook identifies *seven* interrelated factors: speed and volume of communication, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. It does not present an agent-based generative model, a semantic misinformation scale, or a tipping-time estimator.

This point prevents a common attribution error. The six simulation dimensions used in this thesis—valence, surprise, identity alignment, network clustering, information echo, and temporal acceleration—are a *computational remapping*, not Pfeffer et al.’s original list. Valence represents the affective character of firestorm messages. Identity alignment and information echo approximate lack of diversity and clustered reinforcement. Network clustering maps directly to local structure, while surprise and temporal acceleration separate seeding shocks from the speed of subsequent exchange. Cross-media dynamics has no equivalent in the present single-platform society and is a boundary condition. Nor does the simulator reproduce the original paper’s binary platform affordances in a literal way. The remapping is thus an operational hypothesis to be assessed, not a claim that all seven mechanisms have been implemented.

Later work has moved toward measuring observed firestorms. Nuortimo, Karvonen, and Härkönen (2020) propose a three-level scale based on a media-analytics case, and Nuortimo et al. (2025) develop dimensions of firestorm width, intensity, and peak duration with AI-assisted validation. These studies make event severity more explicit, but remain different from a claim-level instrument applied during a controlled generative cascade. The literature leaves open how a theory of rapid collective outrage should be connected to the semantic mutation of scientific information.

2.3 Geoengineering discourse as the substantive case

Solar geoengineering is an unusually demanding case for studying that connection. Discussion combines scientific uncertainty, climate politics, governance, distrust, and established conspiracy narratives. It therefore cannot be represented adequately as generic “fake news.” The principal empirical grounding is Debnath et al. (2023b), who analysed 814,924 English-language tweets containing *#geoengineering* from 2009 to 2021. Using natural-language processing, toxicity analysis, word embeddings, and network analysis, they show that the false chemtrails narrative substantially shaped geoengineering discussion. Conspiracies spilled into debates in the United Kingdom, United States, India, and Sweden and connected with broader political themes. Positive emotions tended to increase following governance events, whereas negative and neutral emotions increased around project and experiment announcements; toxicity widened anti-solar-radiation-management spillovers.

Debnath et al. establish an observational association, not a causal diffusion model or an LLM-agent society. Their corpus is nevertheless valuable for defining plausible discourse tendencies and seed events. This thesis derives reduced belief profiles for conspiracy, climate-action, and environmental-concern positions from that account. These profiles are *theory-grounded types*; they are not user clusters or fitted population proportions. The public data deposit contains tweet identifiers (Debnath et al., 2023a). The thesis did not hydrate and recluster the 814,924 tweets and therefore does not claim to replicate Debnath et al.’s analysis, infer their social graph, or construct a Twitter digital twin.

This separation supports a defensible use of the case. The observational study supplies the empirical phenomenon—chemtrails-related spillover, affect, and toxicity—while the simulation supplies counterfactual control over seeds, identity composition, and network structure. Agreement between the two would be evidence of external plausibility, not proof that the simulated agents are the users in Debnath et al.’s data.

2.4 Network topology, homophily, and echo

Topology changes who can expose whom and how many independent or reinforcing paths connect groups. Erdős–Rényi graphs provide a random-graph reference (Erdős and Rényi, 1960); Watts–Strogatz networks combine local clustering with short paths (Watts and Strogatz, 1998); and preferential attachment generates heavy-tailed degree distributions and hubs (Barabási and Albert, 1999). These models are not interchangeable proxies for real social media. They are controlled structural hypotheses whose degree distribution, clustering, path length, and seed position must be reported.

The content being transmitted also changes which topology should facilitate diffusion. Centola and Macy (2007) distinguish simple contagion, for which one contact may suffice, from complex contagion requiring reinforcement from multiple neighbours. Long weak ties can accelerate the former but fail to provide sufficiently wide bridges for the latter. Firestorm participation and conspiracy reinterpretation may involve social affirmation and are therefore not safely assumed to behave like infection over a single edge.

Agent composition and topology are joined by homophily. Similarity structures social ties and thereby the information, attitudes, and interactions available to people (McPherson, Smith-Lovin, and Cook, 2001). Empirically, selective exposure and homogeneous interaction can support polarised information communities: Del Vicario et al. (2016) relate emotional contagion and group polarisation in Facebook communities, while Törnberg (2018) models viral misinformation as a complex contagion. A high-level “echo chamber” label, however, is not a measurement. This thesis therefore distinguishes treatment variables (network generator and persona assignment) from observables such as edge homophily and community separation. Modularity compares within-group edge density with a degree-preserving null expectation (Newman, 2006); it indicates structural segregation, not the truth or extremity of group content.

2.5 Homogeneous and heterogeneous agent populations

Homogeneous-agent models isolate a response rule and simplify attribution, but they can turn an imposed persona into a population-wide mechanism. Heterogeneous populations permit cross-cutting exposure, specialist roles, and different propensities to reinterpret or resist a message. At the same time, heterogeneity is not intrinsically realistic. Results depend on which types are included, their prevalence, their network positions, and whether prompts create genuine behavioural variation or merely labels.

The immediate methodological predecessor is the auditor–node system of Maurya et al. (2025), presented at the LASS workshop at ACM CIKM 2025. It compares persona-homogeneous and persona-heterogeneous linear rewrite chains and introduces the discrete Misinformation Index (MI) and branch-level Misinformation Propagation Rate (MPR). Those general-news linear-chain findings are *prior work*, not results of this thesis. The present work retains the auditor–node idea but changes the empirical object to geoengineering discourse, uses Debnath-grounded profiles, and introduces non-chain network treatments.

The broader literature demonstrates both the promise and the unresolved status of LLM societies. Social Simulacra populates prototypes of online communities (Park et al., 2022), while Generative Agents combines memory, reflection, and planning to create believable local interactions (Park et al., 2023). S³ simulates information, attitude, and emotion propagation against social-network data (Gao et al., 2023). OASIS adds platform actions, recommendation mechanisms, dynamic networks, and very large populations (Yang et al., 2024); AgentSociety similarly prioritises large-scale social environments (Piao et al., 2025). These systems expand scale and behavioural surface area, but scale does not by itself validate an agent as a model of a person.

Most directly, Li et al. (2026) condition LLM-agent forwarding on psychological traits and compare diffusion under different network structures, including an empirical Higgs community. Their topology-sensitive forwarding study is close to this thesis, but the focal state remains sharing and diffusion rather than audited claim-by-claim mutation in geoengineering text. The literatures are complementary: one asks whether an item is forwarded; this thesis asks how the item’s factual content changes as a function of identity mix and topology.

Persona conditioning also creates a validity problem. Language models can reproduce some subgroup response patterns when conditioned on demographic backstories (Argyle et al., 2023), and they can recover qualitative patterns from selected economic experiments (Horton, 2023). Such results support LLMs as inexpensive theory probes, not synthetic replacements for human participants. Training-data contamination, prompt sensitivity, stochasticity, stereotype amplification, limited within-group variance, and obedience to experimenter instructions all threaten behavioural interpretation. In this thesis, “homogeneous” and “heterogeneous” consequently describe prompt conditions in a computational experiment. They do not denote representative human populations.

2.6 Measurement validity: discrete and continuous instruments

The thesis’s central dependent variable is produced by an instrument, not observed directly. Construct validity concerns whether evidence supports the interpretation and use of a score. It cannot be established by naming a quantity “misinformation.” Cronbach and Meehl (1955) require a measure to sit in a nomological network connecting a construct to observable consequences, and Messick (1995) treats validity as an integrated argument about score meaning, generalisation, external relations, and consequences. For MI, relevant evidence would include item coverage, human agreement, expected relations with known distortions, stability across judges, and sensitivity to irrelevant wording.

QA-based factuality evaluation supplies a methodological warrant. Fabbri et al. (2022) show that question answering can assess factual consistency in summarisation more directly than lexical overlap. LLM judges can also agree substantially with human preferences, but exhibit position, verbosity, and self-enhancement biases (Zheng et al., 2023). Neither result validates this thesis’s exact instrument: fixed binary questions about geoengineering, answered by an LLM judge, are a different construct and use case.

The distinction between discrete and continuous scoring must therefore remain visible. In discrete MI, each item is classified as recovered, missing, or contradicted and the item outcomes are aggregated into a bounded score. This format is inspectable and corresponds to explicit propositions, but it collapses uncertainty and treats all items as equally diagnostic. A continuous instrument can retain graded evidence or confidence and may detect smaller changes, but its numeric precision is not automatically substantive precision. It requires calibration and a defensible interpretation of distances between scores.

The statistical literature cautions that converting a genuinely continuous quantity into categories discards information, reduces power, and creates artificial discontinuities around a cut point (MacCallum et al., 2002; Altman and Royston, 2006). That warning applies to the thesis’s severity bands: the boundary at $MI > 3$ is an operational decision inherited from prior work, not a psychometrically established natural kind. Conversely, the warning does not prove that a continuous LLM score is valid. Discrete and continuous MPR are different instruments; they should not be pooled or treated as interchangeable observations merely because both have numeric output.

Shared generator and judge families add common-method dependence. A fluent rewrite may be rewarded because it resembles the judge’s preferred formulation, and a failed parse can masquerade as low misinformation if the implementation fails open. Credible validation therefore requires frozen test cases, fail-closed parsing, deterministic judging where possible, manual inspection, human coding of a sample, and agreement reported at the decision boundary used for the substantive claim. Until such evidence exists, MI and MPR are best described as *simulation audit scores*, not validated measures of human misinformation belief.

2.7 Validation of generative social simulation

Verification asks whether the program implements the intended model; validation asks whether the model is adequate for its intended purpose. Sargent (2013) separates data

validity, conceptual-model validity, computerised-model verification, and operational validity. For agent-based social simulation, Richiardi et al. (2006) further stress transparent model description, sensitivity analysis, replication, and a validation test matched to the claim. Windrum, Fagiolo, and Moneta (2007) show that empirical validation can involve several strategies and that no single fitted trajectory settles model adequacy.

Applied here, validation has at least four levels. First, software checks must establish that topology, persona assignment, intervention timing, and auditor aggregation behave as specified. Second, internal validation asks whether changing a theoretically relevant input produces stable, interpretable changes rather than parser artefacts or dead cascades. Third, output validation compares patterns with held-out empirical regularities, such as plausible cascade structure or Debnath’s discourse themes. Fourth, measurement validation asks whether humans and independent instruments support the MI interpretation. Face-plausible agent prose is evidence for none of these levels by itself.

The intended use defines the attainable claim. A small, replicated, audited model may function as a mechanism probe: it can test whether a proposed combination of topology and identity is sufficient to produce a pattern under specified prompts. Without calibrated agent distributions and reconstructed empirical cascades, it cannot estimate the prevalence or causal effect of that mechanism among Twitter users. Failed or near-empty cascades are execution or design outcomes to diagnose, not evidence that a topology prevents misinformation. Stochastic LLM output also requires multiple seeds or models before a directional contrast can be considered stable.

2.8 Synthesis and research gap

The literature leaves a precise intersection underdeveloped. Firestorm theory identifies structural conditions for rapid, affective outrage but does not provide a generative, claim-level model (Pfeffer, Zorbach, and Carley, 2014). Network diffusion research explains activation, reinforcement, and cascade shape but usually treats content as a binary or fixed state (Kempe, Kleinberg, and Tardos, 2003; Centola and Macy, 2007). Debnath et al. provide a large observational map of geoengineering conspiracy spillovers but not controlled counterfactual societies (Debnath et al., 2023b). LLM societies generate heterogeneous behaviour at increasing scale, yet their misinformation evaluations commonly emphasise sharing, aggregate plausibility, or believability rather than a validated trajectory of proposition-level change. Finally, the prior auditor–node study supplies MI/MPR and homogeneous versus heterogeneous chains, but not a geoengineering-grounded network experiment (Maurya et al., 2025).

The gap filled by this thesis is therefore *not* a new general-purpose agent architecture and not a replication of an 814,924-tweet study. It is a theory-mapped, auditable test of how network topology and Debnath-grounded identity composition jointly condition semantic misinformation trajectories in geoengineering discourse. It connects (i) an explicit seven-to-six Pfeffer operationalisation, (ii) controlled homogeneous and heterogeneous LLM-agent societies, and (iii) discrete and continuous claim-recovery instruments whose validity limits remain explicit. The empirical contribution is the resulting climate-domain comparison and its

validation evidence; the methodological contribution is a separation of transmission, semantic mutation, firestorm timing, and instrument choice. This design addresses a question not answered by any one parent literature: under fixed scientific seeds, are apparent firestorm thresholds robust to who occupies the network, how those agents are connected, and how factual degradation is measured?

3 Theoretical Framework and Operationalisation

3.1 Conceptual framework

3.1.1 Purpose and analytical levels

This thesis explains how factual content changes while it moves through a synthetic social network. The explanation has three levels. Persona composition defines the agents that transform a message. Topology defines who can expose whom. Propagation joins these elements over time. The resulting texts are then scored with an explicit measurement instrument.

This sequence separates mechanism from measurement. The simulated mechanism produces texts, actions, and exposure paths. The auditor converts those texts into misinformation scores. The scores are observations of the simulation. They are not the mechanism itself.

The framework also separates misinformation from a firestorm. Misinformation concerns departure from factual source claims. An online firestorm concerns a rapid and large discharge of negative word-of-mouth (Pfeffer, Zorbach, and Carley, 2014). The two phenomena may coincide. They need not coincide. Accurate criticism can form a firestorm. Severe misinformation can also diffuse slowly and quietly. Therefore, Misinformation Index (MI) and Misinformation Propagation Rate (MPR) do not by themselves identify a firestorm.

Three kinds of statement follow from this distinction. *Theory* states a proposed relation among composition, topology, exposure, and content change. *Operationalization* states how the relation is represented in the simulation. *Evidence* states what an observed source or experiment can support. The distinction is maintained throughout this chapter.

3.1.2 The integrated causal argument

Let $G = (V, E, W)$ be a directed network. The node set V contains synthetic agents. The edge set E defines possible transmissions. The weights W represent relational conditions such as trust. Each node i receives a persona p_i . The complete assignment is denoted by $\mathbf{p} = (p_1, \dots, p_{|V|})$.

Let X_e be the text carried by propagation event e . When node i processes the event at time t , it applies a persona-conditioned transformation:

$$X_{e'} = T_{p_i}(X_e, H_i(t), R_i(t)). \quad (3.1)$$

Here, $H_i(t)$ is the information available to the agent. $R_i(t)$ denotes local relational conditions. The output may be forwarded, reinterpreted, or dropped. Thus, propagation is a sequence of selective text transformations. It is not the movement of a fixed information token.

3 Theoretical Framework and Operationalisation

The theoretical object is the joint process

$$\mathcal{Y} = F(\mathbf{p}, G, S, A, \mathcal{T}, \varepsilon), \quad (3.2)$$

where S is the seed message. A is the action rule. $\mathcal{T}$ is the exposure schedule. ε collects stochastic and model-specific variation. $\mathcal{Y}$ contains generated texts, transmissions, and cascade structure. An auditor J then maps the generated texts to observed scores:

$$\widehat{\mathcal{M}} = J(\mathcal{Y}; Q, G_Q, \theta_J). \quad (3.3)$$

The question set Q , gold answers G_Q , and auditor specification θ_J are part of the measurement model. They are not neutral bookkeeping choices.

The framework therefore has a clear order. Composition and topology shape exposure. Exposure selects which persona transformations occur. Repeated transformations produce factual retention or drift. The auditor then measures that drift. Pfeffer’s firestorm theory interprets the broader communication conditions. Debnath’s belief profiles ground the identities and issue domain.

3.1.3 Persona composition as a transformation mechanism

Theory. A persona is a structured prior over attention, interpretation, and expression. It makes some source claims salient. It makes other claims easy to omit or recode. Repeated use of the same prior can compound a stable distortion. Mixed priors can interrupt that process. They can also introduce new distortions. Heterogeneity is therefore not intrinsically corrective.

Composition has at least three dimensions. The first is the share of each persona family. The second is the diversity of those families. The third is their placement in the graph. Two societies can have the same shares but different outcomes. One may place a conspiracy persona at a hub. The other may place the same persona at the periphery. The label “heterogeneous” does not capture this difference.

This point follows directly from the award paper’s results. In homogeneous 30-node branches, identity- and ideology-laden personas often accelerated drift. Expert and neutral personas often stabilized factual content (Maurya et al., 2025). Yet the controlled-random heterogeneous branches reached the propaganda tier in about 85% of branch–domain pairs. The prior study therefore does not establish a general “diversity buffers misinformation” law. It shows that persona effects depend on order, content, and repeated interaction.

Operationalization. The thesis assigns persona identifiers and prompts to nodes. Homogeneous conditions repeat a focal identity. Heterogeneous conditions combine several identities. The relevant treatment is the complete assignment $\mathbf{p}$, not the H/He label alone. Composition should therefore be reported as persona-family shares and node positions. Results should also be conditioned on persona family. A lower network mean may otherwise reflect fewer conspiracy seats rather than corrective interaction.

Evidence. The award paper provides simulation evidence for persona-conditioned drift on linear branches. It does not provide human behavioural validation. It also does not provide evidence about complex-network placement. The present experiments can provide evidence about synthetic agents under fixed prompts and models. They cannot establish that real users with similar labels would behave identically.

3.1.4 Topology as an exposure mechanism

Theory. Topology does not directly alter a sentence. It changes the opportunities for alteration. Degree controls how many transmissions can follow an event. Path length controls how many sequential transformations can occur. Clustering creates repeated exposure from connected neighbours. Community structure concentrates exposure within groups. Bridges permit a frame to cross between groups. Hubs place selected agents in high-leverage positions.

Topology and composition are therefore non-separable. A clustered graph does not create an identity echo unless identities align with its clusters. A homogeneous graph produces perfect identity agreement by construction. That agreement is not evidence of an emergent echo chamber. Likewise, a scale-free graph is primarily a hub contrast. It is not automatically a clustering contrast.

Operationalization. The topology generator defines E and initial edge conditions. Persona assignment then defines the relation between social position and identity. The design includes chains, rings, random graphs, small-world graphs, scale-free graphs, echo chambers, polarized graphs, and hierarchical graphs. Each topology represents a distinct exposure opportunity structure. No generator is treated as a complete model of a platform.

The linear chain is the direct link to the award paper. The small-world graph is the clearest named operationalization of local clustering (Watts and Strogatz, 1998). The scale-free graph represents unequal connectivity and hub exposure (Barabási and Albert, 1999). The random graph provides a stochastic baseline (Erdős and Rényi, 1960). The echo and polarized graphs encode pre-existing segregation. They do not demonstrate that segregation emerged during the run. The hierarchical graph represents asymmetric transmission. It does not operationalize legacy-media feedback.

Evidence. Differences between topology cells are evidence about the implemented generators. They are not evidence about all networks in the same named class. One graph seed per cell does not identify a topology-wide effect. Dead cascades also carry information about reachability. They must not be recoded as zero misinformation.

3.1.5 Propagation as a coupled selection process

Theory. Propagation determines which potential transformations become realized. At each event, an agent may forward, reinterpret, or drop a message. Forwarding preserves more of the received text. Reinterpretation invokes another persona-conditioned rewrite. Dropping ends that path. The observed cascade is therefore selected by both network access and agent action.

This creates two distinct margins. The extensive margin concerns whether content reaches another node. The intensive margin concerns how much the content changes when it is processed. A topology can increase reach without increasing factual drift per rewrite. A persona can increase drift while reducing onward transmission. These mechanisms should not be collapsed into one mean.

Repeated exposure adds a further mechanism. Several neighbours may deliver related frames to the same node. The node can then receive social confirmation as well as content. Pfeffer describes this local echo as a consequence of network clusters (Pfeffer, Zorbach, and Carley, 2014). In the simulation, however, repeated receipt is an implemented exposure pattern. It is not direct evidence of human persuasion.

Operationalization. The simulator records event time, hop count, sender, receiver, action, and rewritten text. Ticks and hops answer different questions. A tick is a simulation update. A hop is a position along a transmission path. Neither is a unit of wall-clock time. The eight-tick horizon is a compressed experimental horizon. It is not an estimate of a Twitter half-life. It is also not Debnath’s eight-word skip-gram window.

The action set is ternary. It includes forwarding, reinterpretation, and dropping. This differs from Pfeffer’s binary-choice factor. No result from the ternary mechanism can confirm a binary-choice proposition without a dedicated ablation.

Evidence. Propagation logs support claims about simulated reach, depth, breadth, and rewriting. They do not support claims about platform speed in minutes or hours. They also do not identify causal persuasion by repeated exposure. Such claims would require a matched manipulation of exposure while other conditions remain fixed.

3.1.6 MI and MPR as a measurement model

Theory. Factual drift is latent until a measurement rule is applied. The auditor–node framework makes that rule explicit (Maurya et al., 2025). The source article defines a set of factual questions. The auditor evaluates whether each answer remains recoverable from a rewrite. MI summarizes deviation at one scored event. MPR summarizes MI over a specified collection of events.

In the award paper, the answer vector has ten binary items. MI is the Hamming count between the reference and audited vectors. Branch MPR is the mean MI across a 30-node branch. Despite its name, MPR is not a temporal derivative. It is a branch-level average severity measure. This matters when the concept is transferred to a graph.

Operationalization. For discrete scoring with m questions, let $z_{ej} = 1$ when answer j is not preserved at event e . Then

$$\text{MI}_e^{(d)} = \sum_{j=1}^m z_{ej}. \quad (3.4)$$

The present climate instrument uses $m = 5$. Its numerical range therefore differs from the award paper’s ten-item instrument.

On a graph, there is no unique branch average. The aggregation set must be named. A node summary can be written as

$$\text{MPR}_i = \frac{1}{|E_i^{\text{scored}}|} \sum_{e \in E_i^{\text{scored}}} \text{MI}_e. \quad (3.5)$$

A network-time summary can instead average scored events at tick t :

$$\overline{\text{MI}}(t) = \frac{1}{|E_t^{\text{scored}}|} \sum_{e \in E_t^{\text{scored}}} \text{MI}_e. \quad (3.6)$$

These quantities answer different questions. Neither should be labelled simply “MPR” without its aggregation rule.

The propaganda threshold is also operational. The award paper labels averages at or below 1 as factual error. It labels values above 1 and at or below 3 as lies. It labels values above 3 as propaganda (Maurya et al., 2025). The thesis reuses these cuts. They are design thresholds. They are not validated psychometric boundaries.

Evidence. MI supports a claim about auditor-detected recoverability of selected source claims. It does not directly measure belief, intent, toxicity, affect, or public harm. MPR supports a claim about the average of those event scores over a declared set. It does not show that misinformation propagated faster. The award paper assumes a perfect auditor in its limitations. The present thesis must instead treat auditor error as a validity threat.

3.1.7 Instrument dependence

Theory. Observed misinformation is jointly produced by the generated text and the instrument. Formally,

$$\widehat{M}_{e,r} = J_r(X_e, Q, G_Q), \quad (3.7)$$

where r indexes an auditor mode. Changing r can change levels, rankings, and threshold crossings. The latent text X_e may remain unchanged. Therefore, a score difference across instruments is not automatically a content difference.

Operationalization. The continuous campaign and the dual-discrete campaign are separate measurement regimes. The continuous auditor returns graded claim-level assessments. The discrete headline counts claim failures. The continuous sidecar in a dual run is a diagnostic attached to that regime. It is not a third ground truth. Scores from different regimes must not be averaged. Threshold crossings must be computed within one regime.

Instrument dependence has four sources. First, question selection determines which facts can be lost. Second, response coding determines whether partial preservation receives credit. Third, the auditor model introduces judgement variance and bias. Fourth, aggregation determines which nodes and events receive weight. Topology can affect the fourth source by changing event counts. An event-weighted mean can therefore partly reflect cascade selection.

Evidence. Agreement across instruments strengthens robustness. Disagreement identifies a measurement sensitivity. It does not reveal which instrument is true. External validation requires human coding or another justified reference. LLM-as-judge studies motivate automated assessment, but they do not validate this climate-specific instrument by transfer alone (Zheng et al., 2023).

3.1.8 From branches to complex networks

Theory. The award paper studies sequential transmission chains. Each node has one predecessor in the realized branch. Depth and exposure order are therefore aligned. A complex network breaks this alignment. A node can receive several messages. A message can follow several paths. Communities, bridges, hubs, and cycles can alter both reach and rewrite order.

The extension is conceptual, not merely technical. In a chain, composition is an ordered list. In a graph, composition becomes a joint distribution over identities and positions. In a chain, MPR has one natural branch denominator. In a graph, several valid denominators exist. In a chain, a late node implies many prior rewrites. In a graph, a late tick need not imply a long path.

The generalization therefore preserves the event-level MI concept. It does not preserve every branch-level interpretation. Graph analyses must report path, node, event, and time aggregation separately.

Operationalization. The chain serves as a bridge to prior work. The topology grid then changes the adjacency structure while retaining the same basic rewrite and audit process. Persona composition is crossed with topology where the design permits. This crossing tests whether identity effects depend on exposure structure. It does not constitute a literal replication of the 30-hop study. The climate articles, persona inventory, model, horizon, and graph process differ.

Comparisons also require common support. Reach, event count, and dead-cell status should accompany MI summaries. A live-only mean describes surviving cascades. It does not estimate the mean over cascades that failed to propagate. Similarly, a 63-node imported graph should not be pooled with an eight-node generated graph. The opportunity structures differ too sharply.

Evidence. The complex-network experiments can show whether a prior pattern survives a domain and structure shift. They cannot reproduce the award paper merely by reusing MI and MPR. Evidence of generalization requires directional stability under the changed object. Evidence of failure is also informative. It places a boundary on the chain result.

3.1.9 Debnath belief profiles as empirical grounding

Theory. Debnath et al. describe geoengineering discourse as an identity-structured information environment (Debnath et al., 2023b). Chemtrails conspiracy content dominates the observed corpus. Smaller streams concern climate action and environmental risk. These

streams provide a domain-specific basis for synthetic belief profiles. They also motivate identity alignment between a message and a persona.

The belief profiles are mechanisms of selective interpretation. A conspiracy profile is expected to assimilate ambiguous geoengineering information into a chemtrails frame. A climate-action profile may foreground governance and justice. An environmental profile may foreground ecological risk. Added scientist, journalist, and moderation roles supply institutional accuracy norms. These added roles support the experimental contrast. They are not empirical Debnath clusters.

Operationalization. The persona library translates discourse types into prompts, tags, tones, and expertise fields. This is a theory-faithful reduction. It is not a reconstruction of individual Twitter users. The full tweet corpus was not hydrated. HDBSCAN profiles were not re-estimated. The imported 63-node structure is a hashtag co-occurrence graph. It is not a retweet cascade.

Debnath’s skip-gram context window concerns words inside tweets. It does not define simulation hops. Debnath also reports no empirical MI or MPR. Twitter toxicity, semantic proximity, and simulated factual drift remain different constructs.

Evidence. Debnath provides observational evidence about the issue domain, discourse streams, and platform corpus. It supports the content validity of the persona families. It does not validate their exact prompts. It also does not validate simulated propagation rates. Claims about the synthetic profiles must therefore remain claims about grounded experimental agents.

3.1.10 Pfeffer firestorms as the macro-theoretical frame

Theory. Pfeffer, Zorbach, and Carley define online firestorms as sudden, large discharges of negative word-of-mouth (Pfeffer, Zorbach, and Carley, 2014). Their Outlook identifies seven interrelated factors:

1. speed and volume of communication;
2. binary choices;
3. network clusters;
4. unrestrained information flow;
5. lack of diversity;
6. cross-media dynamics; and
7. network-triggered decision processes.

This list is the theoretical source. It is not a list of simulator parameters.

The framework maps these factors onto different analytical levels. Speed and volume concern propagation intensity. Binary choices concern the action architecture. Clusters and

Table 3.1: Communication escalation and factual degradation are distinct outcomes.

	Low factual degradation	High factual degradation
Low communication escalation	Limited, largely faithful diffusion	Localized misinformation without a firestorm
High communication escalation	Viral but factually stable communication; it may still be an affective firestorm	A misinformation-amplifying firestorm candidate, if negative affect is also present

unrestrained flow concern topology. Lack of diversity concerns composition and homophily. Network-triggered decisions concern repeated social exposure. Cross-media dynamics concerns an external feedback layer. The affective character of firestorms belongs to the definition. It is not an eighth numbered factor.

Operationalization. The thesis uses a computational remapping. Identity composition operationalizes part of lack of diversity. Graph generators operationalize selected cluster and flow conditions. Seed fan-out can operationalize shock versus drip. Ticks and activity schedules can operationalize simulated opportunity rates. These mappings are partial.

Two factors remain especially constrained. The action architecture is ternary rather than binary. The model has no legacy-media broadcast layer. Cross-media dynamics is therefore held outside the simulation. The hierarchical topology cannot substitute for that missing layer.

Valence, surprise, identity, clustering, echo, and temporal acceleration are thesis-level knobs. They are not Pfeffer’s original factor names. Their value lies in making parts of the theory manipulable. Their limitation lies in incomplete construct coverage.

Evidence. Pfeffer’s article provides qualitative and case-based theoretical evidence. It does not provide an LLM model, MI, MPR, or a numerical tipping threshold. An MI value above 3 is therefore not a Pfeffer firestorm criterion. Simulation evidence can test patterns implied by the remapping. It cannot confirm all seven factors when some factors are held or absent.

3.1.11 A two-axis outcome model

The integrated framework uses two outcome axes. The first axis is communication escalation. It includes reach, volume, breadth, depth, and simulated speed. The second axis is content degradation. It includes event MI and declared MPR aggregates. Firestorm interpretation also requires negative or affective content.

This yields four conceptual cases.

The final cell is the thesis’s central concern. It is still a candidate category. The simulation must show escalation and degradation separately. It must also justify the affective interpretation. No single mean MI can establish all three conditions.

3.1.12 Derived propositions

The framework yields five propositions. They organize the empirical analysis. They do not report findings.

P1: composition–alignment proposition. Factual drift should increase when a persona’s interpretive prior aligns with a misinformation frame. The effect should be strongest under repeated use of that persona. However, heterogeneous composition may either interrupt or compound drift. Its direction depends on order and role composition.

P2: position–composition proposition. The effect of a persona depends on network position. Placing an aligned persona at a hub should alter more downstream opportunities than placing it at the periphery. Composition shares alone are therefore insufficient.

P3: cluster–echo proposition. Clustering should increase repeated local exposure. Identity echo should increase only when persona placement aligns with the clustered structure. Structural clustering and identity homophily are distinct conditions.

P4: reach–severity proposition. Topology may change cascade reach without changing drift per scored rewrite. Persona transformations may change drift without increasing reach. Communication escalation and content degradation should therefore be analysed separately.

P5: instrument–conditional proposition. The magnitude and threshold status of measured drift depend on the auditor regime. A substantive conclusion is robust only when its direction survives justified measurement alternatives. Instrument disagreement is a result about measurement. It is not evidence of two underlying cascades in the same run.

3.1.13 Theory, operationalization, and evidence ledger

3.1.14 Scope of inference

The framework supports mechanism-oriented simulation claims. It can show that a specified persona, graph, seed, and auditor produce a specified pattern. It can compare matched cells within one measurement regime. It can test whether chain patterns persist on more complex networks.

The framework does not support direct population inference. The agents are not sampled humans. The graphs are not representative platform samples. The auditor is not an error-free observer. The imported hashtag graph is not a reconstructed diffusion cascade. Accordingly, the evidence is computational and conditional.

This limitation does not remove the theoretical contribution. It clarifies it. The contribution is an explicit bridge across three literatures. Pfeffer supplies the macro conditions of online firestorms. Debnath supplies the climate-conspiracy identities and domain. Maurya et al. supply the persona rewrite and QA-auditor instrument. The thesis joins these elements in a complex-network design. It then tests where the bridge holds, where it depends on measurement, and where it fails.

Table 3.2: Separation of theoretical constructs, operational choices, and admissible evidence.

Construct	Theory	Operationalization	Evidence boundary
Persona composition	Distribution and placement of interpretive priors shape text transformation.	Persona prompts; homogeneous and heterogeneous assignments; family shares; node positions.	Synthetic-agent evidence only. H/He alone does not identify diversity effects.
Topology	Adjacency, clustering, hubs, and bridges shape exposure opportunities.	Eight graph generators plus an imported custom graph.	Generator-specific evidence. One graph seed does not identify a topology class.
Propagation	Selective forwarding, rewriting, and dropping jointly determine realized paths.	Ternary actions; ticks; hops; trust; inbox and horizon constraints.	Simulation time is not wall-clock time. Ternary actions do not test binary choice.
Factual drift	Source claims may be omitted, contradicted, or transformed.	QA-based discrete or continuous auditor.	Measures selected claim recoverability, not belief, intent, toxicity, or harm.
MI	Event-level observed factual deviation.	Five-item climate score within a declared auditor mode.	Scale depends on question count, coding, and auditor.
MPR	Aggregate severity over a declared propagation unit.	Branch, node, event, or time mean with an explicit denominator.	Not a physical rate. Graph aggregates are not automatically comparable with chain MPR.
Debnath profiles	Issue-specific identities condition interpretation of geoengineering claims.	Theory-faithful persona prompts and coarse discourse families.	Grounded reductions, not recovered user clusters or digital twins.
Pfeffer firestorm	Rapid, high-volume, negative word-of-mouth under seven interrelated conditions.	Partial mapping to composition, topology, seeding, and schedule.	MI/MPR alone cannot identify a firestorm. Cross-media remains untested.

4 System, Data, and Methods

4.1 Prior-work boundary

The co-authored LASS workshop paper introduced the auditor–node architecture, the Misinformation Index (MI), and the Misinformation Propagation Rate (MPR) (Maurya et al., 2025). This thesis uses that system as an inherited instrument. Its individual research layer comprises the climate corpus, the Pfeffer mapping, the Phase 2 factorial design, the composition analysis, and the D-net boundary test. The exact division of implementation and writing among co-authors remains an administrative placeholder on the title pages.

4.2 Simulation design

This study used the repository’s agent-based information-diffusion engine to compare the propagation of climate and geoengineering claims across network structures and persona compositions. The design adapts the homogeneous and heterogeneous conditions of the CIKM/LASS simulation study (Maurya et al., 2025); it does not reproduce that study’s crime-news corpus. The present campaign instead uses climate-specific articles and belief profiles. All values reported in this section are taken from the executed Phase 2 configurations, engine source, campaign plan, hand-off record, and raw run schemas.

4.2.1 Experimental units and factorial design

The principal experimental unit was one configuration–article cell. The factorial grid crossed eight network topologies with either 12 homogeneous persona conditions or six heterogeneous persona mixes, two scoring modes, and six seed articles:

$$\underbrace{8 \times 12 \times 2 \times 6}_{\text{homogeneous}} + \underbrace{8 \times 6 \times 2 \times 6}_{\text{heterogeneous}} = 1,728 \text{ cells.} \quad (4.1)$$

Each configuration was executed once ($N = 1$). Accordingly, article cells within a configuration are repeated observations from one simulation run, not independent campaign replicates. The design supports descriptive comparison of the executed cells but not seed-to-seed uncertainty estimation.

The two scoring modes were continuous-only (T2c) and dual (T2d). In a dual run, the discrete score is the headline score and a continuous score is stored as a sidecar for the same event. The two modes are separate measurement instruments and were not averaged. Phase 1 discrete-only results were likewise not inserted into the Phase 2 grid.

4.2.2 Articles and ground truth

The merged article library contains 24 records, but every cell in the main grid uses the same six campaign articles: `scopex_2017`, `chemtrails_gates_2018_2021`, `sai_geoengineering`, `paris_agreement`, `climate_consensus`, and `polar_bears`. The set spans geoengineering research, a chemtrails conspiracy narrative, and established climate-science claims. Each article record contains an identifier, title, source text, five yes/no audit questions, the corresponding Boolean ground-truth answers, and source notes. These structured question–answer pairs provide the reference against which propagated text is scored. The remaining 18 records in the merged library are not additional observations in the 1,728-cell grid.

Articles were processed sequentially within a configuration. Before each article, its prior history was removed and all node inboxes were flushed. The complete source text was then inserted into the designated seed node at tick zero. No competing article group or intervention was enabled in the Phase 2 configurations considered here.

4.2.3 Persona conditions

The merged persona library contains 17 belief profiles. A profile provides at least an identifier, a display name, and a system prompt; the richer campaign records also contain tags, emotional tone, ideological bias, topic expertise, Debnath-type annotations, stylistic attributes, hashtag repertoires, and provenance. These profiles are theory-informed reductions of discourse types, not HDBSCAN centroids reconstructed from the full Debnath Twitter corpus (Debnath et al., 2023b). Their purpose is controlled role conditioning, not demographic representation.

The homogeneous arm (H) uses 12 profiles, one profile per configuration and the same profile at all eight nodes. They comprise four conspiracy profiles (`conspiracy_believer`, `conspiracy_harp_weather`, `conspiracy_depopulation`, and `conspiracy_climate_piggyback`); three climate-action profiles (`climate_action_advocate`, `climate_justice_youth`, and `mitigation_first_policy`); and five science/environment profiles (`environmental_concern`, `ozone_stratosphere_specialist`, `biodiversity_food_security`, `climate_scientist`, and `science_journalist`).

The heterogeneous arm (He) uses six fixed, ordered mixes of eight identifiers. The mixes vary composition rather than drawing personas at runtime. In particular, `mix_00` contains four conspiracy and four climate/environment profiles, `mix_01` contains two conspiracy and six climate/environment profiles, and `mix_02` contains no conspiracy profile. Mixes `mix_03`–`mix_05` introduce the additional moderator, SCoPEX climate-action, and peripheral-skywatcher profiles recorded in the merged library. Node assignment follows the order in each mix for the sequential generators. Echo-chamber and polarised heterogeneous configurations instead request cluster-based assignment from declared persona pools; unavailable identifiers are filtered by the engine before assignment. Consequently, H versus He is a controlled composition contrast, not a generic test of “diversity” in the abstract.

Table 4.1: Network topologies in the Phase 2 main grid.

Topology	Construction and configured parameters
Linear chain	Directed path $v_i \rightarrow v_{i+1}$, initial edge trust 0.7.
Ring	Directed cycle $v_i \rightarrow v_{(i+1) \bmod 8}$, initial edge trust 0.7.
Random ER	Directed Erdős–Rényi graph with edge probability $p = 0.42$; <code>minSeedOutDegree=2</code> is present in the configuration (Erdős and Rényi, 1960).
Small world	Watts–Strogatz-style directed ring lattice with $k = 4$ and rewiring probability $\beta = 0.1$ (Watts and Strogatz, 1998).
Scale free	Preferential attachment with $m = 2$, beginning from a fully connected seed of $m + 1$ nodes (Barabási and Albert, 1999).
Echo chamber	Two alternating chambers; within-/between-chamber edge probabilities 0.75/0.08, base trusts 0.85/0.15, and minimum seed out-degree 2.
Polarised	Two halves; within-/between-group edge probabilities 0.75/0.05, base trusts 0.88/0.10, and minimum seed out-degree 2.
Hierarchical	Bidirectional rooted 3-ary tree with downward/upward base trusts 0.85/0.30.

4.2.4 Network topologies

Every main-grid graph contains eight directed nodes. The eight topology families and their executed parameters are summarised in Table 4.1. Edge trust is stored on the source node. For generators that use persona homophily, the source code adjusts a base trust according to the Jaccard overlap of persona tags and clips the result to $[0.05, 0.95]$.

The linear chain is retained as the direct structural link to the earlier CIKM/LASS design. The other generators test cyclic, random, clustered, hub-dominated, ideologically partitioned, and authority–audience structures. These small graphs are experimental mechanisms; they are not claimed to be fitted reconstructions of a social-media platform.

4.2.5 Seeding and propagation protocol

Each article is drip-seeded into `node_0` only. The seed message has source `ORIGIN`, hop count zero, tick zero, the unmodified article text, and an empty provenance chain. Nodes are always active and have an inbox cap of four messages per article. Processing then proceeds synchronously for at most eight ticks. Messages produced during tick t are delivered only after all nodes have completed that tick and are therefore available from the next tick. The simulation stops early when a tick produces no outgoing message.

For a received message x , the node first rejects the message if its hop count has reached the configured limit or if source trust is below the node’s threshold. It then samples one of three configured actions:

$$A(x) \in \{\text{forward}, \text{reinterpret}, \text{drop}\}.$$

Linear-chain configurations use

$$P(A) = (0.10, 0.85, 0.05), \quad \tau = 0.08, \quad (4.2)$$

where the tuple is ordered as forward, reinterpret, and drop, and τ is the trust threshold. The

Input: one configuration and the six fixed seed articles.

1. For each article a , clear inboxes and prior history for a , then deliver a to `node_0` with $h = 0, t = 0$.
2. For $t = 1, \dots, 8$, process every message at every active node. If $h \geq 8$ or source trust is below τ_v , record DROP. Otherwise sample $A \in \{\text{FORWARD}, \text{REINTERPRET}, \text{DROP}\}$. For REINTERPRET, rewrite the message using the node persona and `gpt-4o-mini`. For either non-drop action, queue the resulting text for all out-neighbours with $h \leftarrow h + 1$.
3. Deliver all queued messages synchronously. Stop early if the queue is empty; otherwise continue with the next tick.
4. After propagation of all six articles, score every forward or reinterpret output against its article’s five ground-truth items.

Figure 4.1: Pseudocode for propagation and post-hoc auditing of one configuration.

other main-grid configurations use

$$P(A) = (0.35, 0.50, 0.15), \quad \tau = 0.15. \quad (4.3)$$

A forward action preserves the received content. A reinterpret action sends the content to the node’s persona-conditioned language model with an instruction to rewrite it in 100–200 words and return only the rewritten text. A drop action records no outgoing text. Forwarded and reinterpreted messages are broadcast to all of the node’s outgoing neighbours with hop count incremented by one.

The configuration includes `relationEvolution=true` and `trustDelta=0.05`. In the executed source, however, propagation of all six articles is completed before the audit phase begins. The auditor then raises or lowers stored source trust by 0.05 according to whether the event’s misinformation index is at most or above 3. Thus these post-hoc trust updates cannot alter the already-completed propagation trajectories in the same run. Belief-state updates, frame analysis, strategic-agent behaviour, network evolution, opinion dynamics, institutional trust, competitive groups, and interventions are disabled in the raw Phase 2 metadata.

4.2.6 Agent and auditor models

Both rewriting agents and the auditor use `gpt-4o-mini`. The agent is conditioned by the assigned persona’s system prompt. The auditor receives the propagated text, each of the five article questions, and the expected yes/no answers. Scores are therefore LLM-as-judge measurements rather than human annotations (Zheng et al., 2023). Auditing occurs after propagation and only for events whose action is forward or reinterpret and whose output text is present.

For discrete scoring, question j receives $d_j \in \{-1, 0, 1\}$, denoting contradiction/distortion, missing information, or a correct statement, respectively. For $m = 5$ questions, the discrete misinformation index is

$$\text{MI}_{\text{disc}} = \sum_{j=1}^m \mathbb{I}(d_j \leq 0) = n_{\text{missing}} + n_{\text{incorrect}}, \quad \text{MI}_{\text{disc}} \in \{0, \dots, 5\}. \quad (4.4)$$

For continuous scoring, the auditor returns $c_j \in [0, 1]$, where 1 denotes fully correct, 0.5 missing or neutral information, and 0 contradiction. The implemented score is

$$\bar{c} = \frac{1}{m} \sum_{j=1}^m c_j, \quad \text{MI}_{\text{cont}} = m(1 - \bar{c}). \quad (4.5)$$

In dual mode, the discrete and continuous auditor calls are issued in parallel for the same text. The event’s top-level `misinfoIndex` is MI_{disc} . The raw event additionally stores both score objects, the absolute diagnostic gap

$$\Delta_{\text{dual}} = |\text{MI}_{\text{disc}} - \text{MI}_{\text{cont}}|, \quad (4.6)$$

and Pearson agreement after mapping the discrete item scores from $\{-1, 0, 1\}$ to $\{0, 0.5, 1\}$. The gap is a measurement diagnostic, not a third misinformation scale.

4.2.7 Eight-hop horizon and computational budget

The CIKM/LASS chain used 30 positions or hops (Maurya et al., 2025). Phase 2 limits each graph to eight nodes, eight ticks, and a maximum of eight message hops. This is an explicit computational-cost reduction: propagation can require an LLM rewrite for each reinterpretation, followed by one auditor call per scored event in continuous mode and two in dual mode. Dense and cyclic graphs can generate many events per tick, so retaining the 30-step horizon while expanding to eight topologies, 288 configurations, six articles, and dual auditing would multiply paid model calls. The eight-hop horizon therefore prioritises factorial coverage over long-tail cascade depth.

This truncation changes the estimand. It measures early diffusion and semantic change within a short, common budget; it does not establish that a cascade has converged, and it weakens any claim of irreversible late-stage propagation. The number eight is also unrelated to Debnath et al.’s eight-word skip-gram context window (Debnath et al., 2023b). The latter is a word-embedding hyperparameter, not a social-network hop protocol.

4.2.8 Raw records and derived cell summaries

Each timestamped run directory contains `metadata.json`, `state.json`, `graph_topology.json`, one `results_<articleId>.json` file per article, a `human_eval_template.csv`, and one state file per node. The raw schemas were used as follows:

metadata.json run status, fully merged configuration, model-usage counts, and per-article result pointers;

state.json propagation and audit checkpoints, completed article lists, current tick, errors, and last-update time;

graph_topology.json realised node labels, directed edges, and initial edge trusts;

nodes/node_*.json node and persona identifiers, model, relations, node parameters, inbox, action counters, and event history;

results_<articleId>.json node summaries and aggregate network, provenance, frame, and IFD metrics.

An event history record contains the tick, article, source node, hop count, action, input and output text, misinformation index, reason, provenance, timestamp, and—when auditing succeeds—the IFD score object. This event-level record, rather than the aggregate result file, is the parser’s source for Phase 2 headline scores.

For article a , let E_a^* be the set of history events with a non-null top-level misinformation index. The parsed cell headline is the event-weighted mean

$$\overline{\text{MI}}_a = \frac{1}{|E_a^*|} \sum_{e \in E_a^*} \text{MI}_e. \quad (4.7)$$

The parser also reports a mean of node-level means. It is retained as a separate quantity because it weights nodes equally rather than weighting all events equally. A cell with at most one scored event is marked dead and hatched after a real LLM run; it is retained in the ledger, excluded from live means, and never imputed as zero.

4.2.9 Reproducibility and limitations

The campaign records **graphRandomSeed**: 42, all merged configurations, the realised topology, node histories, checkpoints, model-usage totals, and the complete parser inputs. The graph seed is consumed by the echo-chamber and polarised builders. Exact replay must nevertheless not be claimed. In the checked source revision, the ER, small-world, and scale-free builders, as well as action sampling, call JavaScript’s unseeded **Math.random**; the ER builder does not consume the additional seeded-options argument passed by the simulation wrapper. Language-model generation and judging can also vary across calls. Together with $N = 1$, these facts make the archived realised graphs and event logs the reproducible record of what was executed, while a fresh run is a protocol replication rather than a bitwise reproduction.

4.2.10 Campaign isolation

Phase 2 was isolated at both configuration and output level. New configurations reside in **thesisExperiment/configs/phase2/**; raw runs in **thesisExperiment/runs_phase2/**; and parsed outputs in **thesisExperiment/results_phase2/**. The Phase 1 directories **thesisExperiment/runs/**, **thesisExperiment/results/tables/**, and **thesisExperiment/configs/full/** were designated read-only and were not overwritten. Parser selection is by experiment name and retains the latest complete timestamped run with positive LLM-call usage. This separation prevents the earlier discrete 12×12 campaign, probes, and custom Debnath-network runs from silently entering the main 1,728-cell factorial analysis.

4.3 Measurement and analysis

4.3.1 Analytical scope and unit of observation

The Phase 2 experiment evaluated how climate and geoengineering messages changed when persona-conditioned agents propagated them through a simulated society. A ground-truth article was introduced at a designated seed node, agents could forward, reinterpret, or drop the message, and an auditor scored the resulting text against five article-specific ground-truth questions. Both the acting agents and the auditor used `gpt-4o-mini`. The resulting scores are therefore LLM-as-judge measurements, not human annotations and not measurements observed on Twitter.

The main factorial grid crossed eight generated topologies with two identity arrangements and two scoring configurations. The topologies were linear chain, ring, Erdős–Rényi, small world, scale free, echo chamber, polarised, and hierarchical. The homogeneous arm (H) placed one of 12 personas throughout an eight-node graph, whereas the heterogeneous arm (He) used one of six eight-persona mixes. Every configuration processed the same six core articles. Runs used eight ticks, eight hops, one graph seed (`graphRandomSeed: 42`), and one replicate. This produced 1,728 article-level cells:

$$(8 \times 12 \times 2 \times 6) + (8 \times 6 \times 2 \times 6) = 1,728.$$

The harvested table contains all 1,728 expected cells and reports no missing cell. The eight-tick budget was a computational cut relative to the 30-hop CIKM setup; it is not Debnath et al.’s “eight”, which refers to a word-embedding context window.

Four nested measurement units must be distinguished:

1. an *item* is one of the five ground-truth questions;
2. a *scored event* is one agent’s audited handling of one article;
3. a *node MPR* is the mean event-level misinformation index for one node and article; and
4. an *article cell* is one configuration–article combination.

The primary cell statistic is `meanMI`, the arithmetic mean of event-level MI over scored events. The parser also records `meanNodeMPR`, the mean of the node-level means. These quantities use the same event-level instrument but weight nodes differently; they are retained as separate columns and are not substituted for one another.

4.3.2 IFD scoring and the MI–MPR hierarchy

The engine labels the auditor record *IFD*. For each event, the record contains a misinformation index (MI), correct, missing, and incorrect rates (CR, MR, and IR), and auxiliary fields. The analysis uses MI as the event score. It does not report hop-wise CR, MR, or IR because those fields are absent from the live analysis tables.

Discrete instrument

For question j , the discrete auditor returns

$$s_j \in \{-1, 0, +1\},$$

where $+1$ denotes a correct answer, 0 a missing answer, and -1 an incorrect answer. With $m = 5$ questions, let n_C, n_M, n_I denote the corresponding counts. The implementation computes

$$\text{CR} = \frac{n_C}{m}, \quad \text{MR} = \frac{n_M}{m}, \quad \text{IR} = \frac{n_I}{m},$$

and

$$\text{MI}_d = n_M + n_I. \quad (4.8)$$

Thus $\text{MI}_d \in \{0, 1, 2, 3, 4, 5\}$. Missing and incorrect answers contribute equally to MI even though the IFD record preserves their separate rates. The dual-scoring campaign, denoted T2d, uses this discrete MI as its top-level **misinfoIndex** and hence as its headline instrument.

Continuous instrument

The continuous auditor returns one score $q_j \in [0, 1]$ per question. Its correctness rate and MI are

$$\text{CR}_c = \frac{1}{m} \sum_{j=1}^m q_j, \quad \text{MI}_c = m(1 - \text{CR}_c). \quad (4.9)$$

For the five-question articles, MI_c is a real-valued score on the same nominal 0–5 range. Scores at least 0.67 are counted as correct and scores at most 0.33 as incorrect when the continuous IFD record derives CR/MR/IR counts; intermediate scores are counted as missing. The continuous-only campaign, denoted T2c, makes one continuous auditor call per scored event and stores MI_c as the event’s headline **misinfoIndex**.

MPR aggregation

For node v , article a , and instrument $r \in \{d, c\}$, the node-level misinformation propagation rate (MPR) is

$$\text{MPR}_{v,a}^{(r)} = \frac{1}{|E_{v,a}|} \sum_{e \in E_{v,a}} \text{MI}_e^{(r)}, \quad (4.10)$$

where $E_{v,a}$ is the set of scored events for that node and article. At cell level, **meanMI** instead averages over all scored events directly:

$$\overline{\text{MI}}_{\text{cell}}^{(r)} = \frac{1}{|E_{\text{cell}}|} \sum_{e \in E_{\text{cell}}} \text{MI}_e^{(r)}. \quad (4.11)$$

The parser’s **meanNodeMPR** is $\frac{1}{|V_{\text{scored}}|} \sum_v \text{MPR}_{v,a}$. It gives each reached node equal weight, whereas **meanMI** gives each scored event equal weight. The chapter therefore uses **meanMI** as the cell headline and presents **meanNodeMPR** only as a companion summary.

4.3.3 Dual scoring and instrument separation

In a T2d event the auditor makes two separate LLM calls in parallel: one discrete call and one continuous call. The returned top-level IFD object mirrors the discrete record. A nested `ifd.dual` sidecar stores the discrete record, the continuous record, an absolute gap, and an agreement coefficient:

$$g_e = |\text{MI}_{d,e} - \text{MI}_{c,e}|, \quad (4.12)$$

$$a_e = \text{cor}\left(\frac{s_{1,e} + 1}{2}, \dots, \frac{s_{m,e} + 1}{2}; q_{1,e}, \dots, q_{m,e}\right). \quad (4.13)$$

The normalisation maps discrete item scores $-1, 0, +1$ to $0, 0.5, 1$, after which the code computes Pearson’s correlation with the continuous item scores. The stored gap is *absolute*; it is not a signed discrete-minus-continuous difference.

This design yields two campaign instruments and one within-event diagnostic:

- T2c headline: continuous MI from continuous-only events;
- T2d headline: discrete MI from dual-run events; and
- T2d sidecar: continuous MI, absolute gap, and item-score agreement on those same dual-run events.

The sidecar continuous score is not the T2c headline because it was obtained in a different campaign call on a different realised event. Likewise, the absolute gap is a scorer-disagreement diagnostic, not a third MPR. Discrete and continuous means are therefore shown in separate columns or panels, never averaged into a common headline. Cross-instrument differences are explicitly labelled as instrument contrasts and cannot identify a change in misinformation while holding the measurement process fixed.

4.3.4 Network threshold k^*

The parser forms a network mean MI at every observed tick t :

$$\overline{\text{MI}}_t = \frac{1}{|E_t|} \sum_{e \in E_t} \text{MI}_e.$$

It then defines

$$k^* = \min \left\{ t : \overline{\text{MI}}_t > 3 \ \wedge \ \overline{\text{MI}}_u > 3 \text{ for every later observed } u \right\}. \quad (4.14)$$

A temporary crossing that later returns to 3 or below does not qualify. If no irreversible crossing is observed, k^* is undefined rather than zero. The observation window ends at eight ticks, so a crossing at the final observed tick is right-censored and provides only weak evidence of persistence. T2d k^* is calculated from the discrete series; T2c k^* from the continuous series. Their rates remain separate. This threshold measures simulated semantic degradation under the study’s MI rule; it is not a Twitter ignition time or an empirical firestorm duration.

4.3.5 Hatch policy and analysis population

A cell is classified as dead when it contains at most one scored event:

$$\text{dead} \iff \text{nScored} \leq 1.$$

If the run nevertheless records positive LLM usage, the parser labels the row `hatched_dead_after_llm`. Such rows remain in the harvest and in the dead-cell ledger, but are hatched in plots and excluded from live-cell means. They are neither deleted nor imputed as $\text{MI} = 0$. Consequently, a dead heterogeneous cell is not evidence that heterogeneity prevented propagation. Missing and in-progress rows, if present, would likewise not be empirical zeros.

The official harvest contains 292 hatched-dead cells out of 1,728 (16.9%): 96 in T2c_H, 98 in T2d_H, 50 in T2c_He, and 48 in T2d_He. The four live populations are therefore 480, 478, 238, and 240 cells, respectively. All reported means, medians, paired-cell contrasts, and exploratory tests use the live subset unless a count is explicitly described as a full-harvest or dead-cell count.

4.3.6 Pre-specified comparison structure

The five comparison families were designed to make the held and varied dimensions visible. A “same topology” comparison means that summaries are aligned by one of the eight topology labels; it does not imply that independently executed T2c and T2d runs share identical event histories.

Table 4.2: Comparison families and admissible interpretation.

ID	Compared slices	Held/varied dimensions	Interpretation rule
C1	T2c_H vs. T2c_He; T2d_H vs. T2d_He	topology and instrument held; identity arrangement varied	H–He contrasts are estimated separately for each instrument.
C2	T2c_H vs. T2d_H	topology and homogeneous arm held; instrument varied	an instrument comparison, not a common-scale treatment effect.
C3	T2c_He vs. T2d_He	topology and heterogeneous arm held; instrument varied	same restriction as C2; dual sidecar is reported separately.
C4	T2c_H vs. T2d_He; T2d_H vs. T2c_He	topology held; identity arrangement and instrument both varied	two labelled, confounded contrasts; no pooled C4 headline.
C5	live cells concatenated within a labelled slice or composition subset; D-net reported separately	cell-level table union across generated topologies	descriptive pooled summaries only; no physical super-graph.

For C1, H and He means are first calculated within topology and mode. Both cell-pooled and topology-equal summaries are retained because unequal live counts can change weighting. C2 and C3 additionally use paired live cells joined on topology, persona or mix, and article. A pair enters only when both corresponding cells are live; a dead cell on either side is never replaced by zero. C4 deliberately crosses identity arrangement and instrument. Its two directions are shown separately because averaging them would manufacture a third, uninterpretable MPR.

4.3.7 Pooled table union versus a physical super-graph

In C5, “pooled” means row concatenation. Let $\mathcal{L}_{r,z,g}$ be the live article-cell rows for instrument r , identity arm z , and topology g . The all-topology pool is the disjoint table union

$$\mathcal{P}_{r,z} = \biguplus_{g=1}^8 \mathcal{L}_{r,z,g}.$$

The pooled mean weights each live article cell equally. The topology-equal companion first calculates one mean per topology and then gives each of the eight topology means weight $1/8$. Reporting both exposes weighting sensitivity.

No edges are added between runs, no node identifiers are merged, and no propagation is rerun on the union. It is therefore incorrect to interpret C5 as a graph with 64 nodes, as a union of adjacency matrices, or as a synthetic super-cascade. The D-net analysis is also not part of this pool. D-net is a separately executed 63-node, 228-directed-edge custom topology based on a hashtag co-occurrence object. The edge container is not a hydrated retweet cascade, and the object has no empirical Twitter MPR.

4.3.8 Descriptive and exploratory statistical procedures

The primary analysis is descriptive. It reports cell counts, live and dead counts, arithmetic means of `meanMI` and `meanNodeMPR`, medians of paired differences where relevant, k^* occurrence rates among live cells, dead-cell rates over all harvested cells, and absolute dual-gap and agreement means from the dual sidecar. Grouped summaries are produced by topology, article, persona or mix, identity arm, and instrument. Heat-map entries with no live cell are left missing, not displayed as zero.

Three non-parametric procedures accompany the descriptive tables:

- two-sided paired Wilcoxon signed-rank tests compare the eight aligned topology means in C1–C3 and, separately, the paired live cells in C2 and C3;
- Spearman correlation describes the rank agreement of topology means between instruments in C2 and C3; and
- two-sided Mann–Whitney U tests compare pooled live H and He cells, conspiracy and science/environment H cells, and `mix_00` and `mix_02` He cells in C5, always within one instrument.

These tests are exploratory. The eight topology means are design strata rather than independent campaign replications, and pooled cells share one graph seed and are nested within topology, persona or mix, and article. With $N = 1$, p -values do not provide confirmatory replicate-level inference. No causal claim, confidence interval, or population-level firestorm effect is inferred from them. The harvest’s `thesisGrade` flag is `false`.

4.3.9 D-net structural validation

Comparable cascade-shape metrics

The validation code compares three structural quantities:

Depth the longest breadth-first-search path from the inferred root;

Breadth the maximum number of reached nodes at any depth level; and

Structural virality the mean pairwise shortest-path distance among reached nodes after treating propagation edges as undirected.

Cascade size is reported for context but excluded from the structural-similarity score. Simulated propagation trees are reconstructed from event provenance for forward and reinterpret actions. The implementation also records simulation span in ticks; the empirical hashtag object has no usable tweet clock, so ticks cannot be converted to hours.

For structural metric $j \in \{\text{depth}, \text{breadth}, \text{virality}\}$, the relative error is

$$e_j = \left| \frac{x_j^{\text{sim}}}{x_j^{\text{real}}} - 1 \right|,$$

and the aggregate structural similarity is

$$S_{\text{struct}} = \max \left(0, 1 - \frac{1}{3} \sum_j e_j \right). \quad (4.15)$$

The implementation additionally marks a metric as a scalar match when $e_j < 0.30$. For D-net, the empirical values were depth 4, breadth 33, and structural virality 2.7; the corresponding means over 16 simulated cascades were 2.5625, 40.9375, and 1.7963. Only breadth met the 30% rule, yielding $S_{\text{struct}} = 0.6885$. These are cascade-shape comparisons, not comparisons of simulated MI with an empirical Twitter MI.

KS and Jensen–Shannon comparisons

For each of depth, breadth, and structural virality, the validation module compares the empirical and simulated value arrays with an approximate two-sample Kolmogorov–Smirnov (KS) test. The KS statistic is the maximum distance between the two empirical cumulative distribution functions. A small statistic indicates closer empirical distributions; the code reports a Kolmogorov-distribution approximation to the p -value.

The same arrays are discretised into 20 common-width bins for Jensen–Shannon (JS) divergence. With base-2 logarithms, JS lies in $[0, 1]$: zero denotes identical binned distributions and one complete separation. The code labels a distributional match only when KS $p > 0.05$ and JS < 0.10 . It defines the aggregate distributional score as

$$D = \max \left(0, 1 - \frac{1}{3} \sum_j \text{JS}_j \right). \quad (4.16)$$

The D-net comparison has one empirical object and 16 simulated cascades. The observed KS statistics were 1.0000 for depth, 0.8125 for breadth, and 1.0000 for structural virality, with approximate p -values 0.1104, 0.2954, and 0.1104. JS divergence was 1.0000 for all three metrics, so no metric met the joint match rule and $D = 0$. Because $n_{\text{real}} = 1$, the KS p -values are severely underpowered and do not establish distributional equivalence. In particular, failure to reject is not evidence that the empirical and simulated distributions match.

Digital Twin Fidelity Score

The code combines structural similarity, distributional score, and a bounded content correlation into the Digital Twin Fidelity Score (DTFS):

$$\text{DTFS} = 0.40S_{\text{struct}} + 0.40D + 0.20\rho_{\text{content}}, \quad (4.17)$$

with each component clipped to $[0, 1]$. The implementation's validation threshold is $\text{DTFS} \geq 0.70$. D-net has $S_{\text{struct}} = 0.6885$, $D = 0$, and $\rho_{\text{content}} = 0$, giving

$$\text{DTFS} = 0.2754 < 0.70.$$

Accordingly, `isValidated` is `false`. This failure concerns the implemented structural/distributional digital-twin check. It does not turn the hashtag graph into a retweet cascade, and it does not license an empirical MPR claim: Debnath's data provide no empirical 0–5 MI/MPR against which the auditor scores could be validated.

4.3.10 Source register and reproducibility boundary

All operational definitions and numerical counts in this chapter are traceable to the following repository artefacts:

- IFD formulas, dual calls, absolute gap, and agreement: `src/Auditor.js`;
- event collection, cell means, node means, k^* , and hatch policy: `thesisExperiment/scripts/parse_phase2.js`;
- harvested counts and the $N = 1/\text{non-thesis-grade}$ status: `thesisExperiment/results_phase2/summary.json`;
- C1–C5 joins, pooled summaries, and exploratory procedures: `thesisExperiment/analysis_full/analyze_full.py` and `thesisExperiment/analysis_full/tables/`;
- structural metric extraction: `src/ValidationMetrics.js`;
- KS, JS, structural similarity, and DTFS definitions: `src/ValidationComparison.js`; and
- D-net validation inputs and outputs: `thesisExperiment/results_phase2/debnath_compare.json`.

Phase 1 tables are not joined to these Phase 2 headlines because they use a different campaign and schema. D-net rows are not concatenated into the 1,728-cell factorial grid. These boundaries preserve the separation among instruments, campaigns, and graph objects throughout the analysis.

5 Results

The results follow the analytical sequence established in the research questions. Instrument sensitivity is reported before persona composition and topology are interpreted. The D-net evidence then tests whether those patterns survive a change from generated eight-node graphs to the separate 63-node complex network.

5.1 Results: MPR instrument contrasts

This section compares the continuous-only campaign (T2c) with the dual campaign’s discrete headline (T2d). The comparison is between instruments, not between two estimates on a common ground-truth scale. For each scored event, the discrete instrument counts the five audit items classified as missing or incorrect and therefore returns an integer from 0 to 5. The continuous instrument combines graded item scores into a floating-point value on approximately the same range. In a dual run, the discrete value is the headline; the continuous value produced for the same event is retained only as a sidecar diagnostic.

All results below are descriptive point estimates from one campaign replicate and eight hops. Dead cells are excluded from means and are never replaced by zero. “Paired” denotes a shared topology–persona/mix–article cell that was live in both separately run campaigns; it does not turn the two auditor outputs into repeated measurements by a single instrument.

5.1.1 C2: homogeneous arm, different instruments

Holding the homogeneous arm fixed, T2d is higher than T2c on all eight topologies (Table 5.1). Averaging the eight topology-specific differences gives

$$\overline{\Delta_H} = \overline{T2d_H - T2c_H} = 0.8201346884.$$

The differences range from 0.6999 on the linear chain to 0.9494 in the echo chamber. There are 398 cells live in both campaigns. Their median paired difference is 0.8125; 342 differences are positive, 26 are negative, and 30 are zero. A further 80 cells are dead only in T2c, 82 only in T2d, and 16 in both, so none of these cells enters the paired difference.

The dual-run diagnostics locate this separation inside the scoring procedure. On the same dual events, the mean absolute gap between the discrete headline and its continuous sidecar is 0.8804 in H, while mean agreement is 0.6487. Agreement is the Pearson correlation between the item-level continuous scores and discrete responses mapped from $\{-1, 0, +1\}$ to $\{0, 0.5, 1\}$. It is therefore a within-event scoring diagnostic, not accuracy against external labels. Likewise, the absolute dual gap is not a third MPR and the sidecar is not the separately generated T2c headline.

Table 5.1: C2 and C3 live-cell headline means by topology. Each Δ is dual-discrete minus continuous within the stated arm. Values are reported to four decimal places from the live full-analysis tables.

Topology	T2c H	T2d H	Δ_H	T2c He	T2d He	Δ_{He}
Linear chain	0.9129	1.6127	0.6999	0.8638	2.0852	1.2214
Ring	0.9160	1.6402	0.7242	0.8760	2.1768	1.3008
Random ER	0.8736	1.7809	0.9073	0.6593	2.0412	1.3818
Small world	0.7215	1.5977	0.8762	0.7545	1.9096	1.1552
Scale-free	0.9205	1.7034	0.7829	0.8157	1.9924	1.1767
Echo chamber	0.8490	1.7984	0.9494	1.3775	1.9632	0.5857
Polarised	0.8356	1.7005	0.8649	1.4137	1.8830	0.4693
Hierarchical	0.8751	1.6315	0.7564	1.5242	2.5934	1.0692
Mean topology Δ			0.8201			1.0450

Table 5.2: Same-event dual diagnostics by topology. Gap is the mean absolute difference between the discrete headline and continuous sidecar; agreement is their item-level response correlation. Neither column is an additional MPR.

Topology	H gap	H agreement	He gap	He agreement
Linear chain	0.9672	0.6223	1.2960	0.7196
Ring	0.8803	0.6656	1.3123	0.6404
Random ER	0.9680	0.6969	1.1335	0.6460
Small world	0.8255	0.5928	1.1049	0.6522
Scale-free	0.8463	0.6607	1.1058	0.6230
Echo chamber	0.8923	0.6306	0.8830	0.5986
Polarised	0.8553	0.6191	0.8784	0.5673
Hierarchical	0.7973	0.6906	1.1354	0.2947
Live-cell summary	0.8804	0.6487	1.1145	0.6174

5.1.2 C3: heterogeneous arm, different instruments

The heterogeneous arm shows the same directional instrument separation: T2d is above T2c on all eight topologies. The topology-equal difference is

$$\overline{\Delta}_{He} = \overline{T2d_{He}} - \overline{T2c_{He}} = 1.0450048645.$$

The topology differences range from 0.4693 on the polarised graph to 1.3818 on random ER. Of the 198 cells live in both campaigns, the median paired difference is 0.97785; 169 differences are positive, 25 are negative, and 4 are zero. Cells dead on one side are again excluded: 42 are dead only in T2c, 40 only in T2d, and 8 in both.

Within the He dual runs, the same-event mean absolute dual gap is 1.1145 and mean agreement is 0.6174. The gap is larger and agreement lower than their H summaries, but these are descriptive diagnostics; with one replicate they do not establish an arm effect. The live tables do not contain CR, MR, IR, or hop-level MI, so no corresponding contrasts are reported.

5.1.3 Why level and rank need not agree

The consistent level ordering does not imply a consistent topology ranking. In H, the topology-mean rank correlation between continuous and discrete headlines is $\rho = 0.0952$; in He it is $\rho = 0.1667$ (eight topology means in each arm). These coefficients are descriptive and are not used as inferential tests.

The apparent combination—T2d above T2c everywhere, but weak agreement in rank—follows from two distinct comparisons. The first asks whether the discrete level exceeds the continuous level within each topology. The second asks whether the size ordering among topologies is preserved. It is not: the instrument increment is topology-dependent rather than a constant offset. For example, the H increment is 0.6999 on the linear chain but 0.9494 in the echo chamber; in He it is 0.4693 on the polarised graph but 1.3818 on random ER. The instruments also use different response representations—hard missing/incorrect counts versus graded scores—and the T2c and T2d campaigns make separate auditor calls. Thus borderline item judgements can change both scale spacing and rank.

These observations explain why “discrete is higher” is an instrument result, not evidence that discrete scoring recovered more true misinformation. Continuous scoring is not a ground-truth benchmark for discrete scoring, and discrete scoring is not a ground-truth correction of continuous scoring. The same-event gaps and non-unit agreement document their divergence; they do not adjudicate which instrument is more valid.

5.1.4 C4: cross-mode, two-factor contrast

C4 changes arm and instrument simultaneously. Comparing T2d He with T2c H gives a positive mean topology difference,

$$\overline{\text{T2d}_{\text{He}} - \text{T2c}_{\text{H}}} = 1.2175517040,$$

whereas comparing T2c He with T2d H gives

$$\overline{\text{T2c}_{\text{He}} - \text{T2d}_{\text{H}}} = -0.6475878488.$$

The sign reverses on every topology when the instrument is swapped against the arm (Table 5.3). This is expected given the large instrument-level separation in C2 and C3. It prevents either cross-mode difference from being interpreted as an H–He effect: each combines identity composition and scoring mode, and neither factor is held fixed.

Accordingly, C4 remains a labelled sensitivity contrast. Averaging its two deltas would cancel differences created by swapping instruments and would produce a quantity with no instrument-specific interpretation. No pooled C4 MPR is therefore reported.

Table 5.3: C4 cross-mode topology means and exact contrast direction. The two deltas change arm and instrument together and are not averaged into a headline.

Topology	T2c H	T2d He	$\Delta(T2d_{He} - T2c_H)$	T2d H	T2c He	$\Delta(T2c_{He} - T2d_H)$
Linear chain	0.9129	2.0852	1.1724	1.6127	0.8638	-0.7489
Ring	0.9160	2.1768	1.2608	1.6402	0.8760	-0.7642
Random ER	0.8736	2.0412	1.1675	1.7809	0.6593	-1.1216
Small world	0.7215	1.9096	1.1881	1.5977	0.7545	-0.8432
Scale-free	0.9205	1.9924	1.0719	1.7034	0.8157	-0.8877
Echo chamber	0.8490	1.9632	1.1141	1.7984	1.3775	-0.4209
Polarised	0.8356	1.8830	1.0474	1.7005	1.4137	-0.2868
Hierarchical	0.8751	2.5934	1.7182	1.6315	1.5242	-0.1074
Mean topology Δ			1.2176			-0.6476

5.2 Persona composition, article valence, and cascade persistence

This section separates the homogeneous-persona (**H**) and heterogeneous-mix (**He**) arms before interpreting them. The labels describe how identities were assigned to the eight nodes: every node received the same persona in H, whereas He used a documented mixture of eight persona identifiers. They do not denote low- and high-diversity treatments with otherwise matched content. In particular, the H arm pools conspiracy personas with climate-action and science/environment personas, while the He arm pools conspiracy-heavy, conspiracy-sparse, and conspiracy-free compositions. Consequently, an aggregate H–He difference combines composition with heterogeneity and cannot by itself identify an effect of diversity.

5.2.1 Homogeneous persona families

Figures 10–12 cross the twelve H personas with the six articles. Figure 10 shows mean misinformation persistence rate (MPR), Figure 11 shows maximum misinformation intensity (MI), and Figure 12 shows the irreversible crossing time k^* . The columns in these heatmaps are not replicates: each cell is one seed, and dead cells are excluded rather than entered as zero.

The strongest separation is by persona family. Across live H cells, the four conspiracy personas have continuous MPR 2.2194 (168 live cells) and dual-discrete MPR 3.1691 (159 live cells). Their k^* rates are 40.5% and 59.7%, respectively. Climate-action personas are much lower: continuous MPR 0.1729 (117 live) and dual-discrete MPR 1.0488 (126 live), with k^* rates 0.0% and 0.8%. The science/environment family is lowest on continuous MPR, 0.1118 (195 live), but its dual-discrete MPR, 0.8687 (193 live), is not zero; its k^* rate is 0.0% in both instruments. Thus the discrete instrument can assign nonzero persistence to science/environment personas without those runs ever locking above the k^* threshold. Figure 20 supports reading this as instrument disagreement: across all live H dual cells the mean discrete-minus-sidecar-continuous gap is 0.8804, and mean agreement is 0.6487.

The individual rows qualify the family averages. On continuous MPR, `conspiracy_believer` (2.4842) and `conspiracy_harp_weather` (2.4829) are nearly tied, whereas `conspiracy_depopulation` (1.9535) and `conspiracy_climate_piggyback`

Table 5.4: Homogeneous personas, collapsed across eight topologies and six articles. MPR means and k^* rates use live cells only; each persona has 48 expected cells per instrument.

Persona	Family	n_c	MPR _c	k_c^*	n_d	MPR _d	k_d^*	Dead c/d
conspiracy_believer	conspiracy	44	2.4842	47.7%	38	3.3522	65.8%	4/10
conspiracy_haarp_weather	conspiracy	40	2.4829	47.5%	44	3.3571	70.5%	8/4
conspiracy_depoulation	conspiracy	43	1.9535	34.9%	37	3.2823	59.5%	5/11
conspiracy_climate_piggyback	conspiracy	41	1.9570	31.7%	40	2.6838	42.5%	7/8
climate_action_advocate	climate action	37	0.2258	0.0%	37	1.0665	0.0%	11/11
climate_justice_youth	climate action	41	0.2454	0.0%	46	1.3072	2.2%	7/2
mitigation_first_policy	climate action	39	0.0464	0.0%	43	0.7571	0.0%	9/5
environmental_concern	science/env.	40	0.1423	0.0%	37	0.8359	0.0%	8/11
biodiversity_food_security	science/env.	39	0.2115	0.0%	37	1.1470	0.0%	9/11
ozone_stratosphere_specialist	science/env.	38	0.1627	0.0%	37	0.9639	0.0%	10/11
science_journalist	science/env.	37	0.0310	0.0%	39	0.6745	0.0%	11/9
climate_scientist	science/env.	41	0.0130	0.0%	43	0.7517	0.0%	7/5

Table 5.5: Heterogeneous mixes collapsed across topologies and articles. Counts and k^* rates refer to live cells; each mix has 48 expected cells per instrument.

Mix	Composition	n_c	MPR _c	k_c^*	n_d	MPR _d	k_d^*	Dead c/d
mix_00	4 conspiracy + 4 climate/env.	39	1.7712	12.8%	40	3.1321	57.5%	9/8
mix_01	2 conspiracy + 6 climate/env.	41	1.4770	9.8%	41	2.7795	58.5%	7/7
mix_02	0 conspiracy + 8 science/climate	41	0.2645	0.0%	41	1.1882	0.0%	7/7
mix_03	1 conspiracy + 7 other	39	0.8945	10.3%	36	2.0410	33.3%	9/12
mix_04	2 conspiracy-adj. + 6 other	38	0.7021	5.3%	39	1.6896	20.5%	10/9
mix_05	2 conspiracy-adj. + 6 other	40	1.0134	7.5%	43	1.6713	23.3%	8/5

(1.9570) form a lower pair. The dual-discrete ordering changes: **conspiracy_haarp_weather** is highest at 3.3571, followed by **conspiracy_believer** at 3.3522, **conspiracy_depoulation** at 3.2823, and **conspiracy_climate_piggyback** at 2.6838. Their dual k^* rates span 42.5%–70.5%; the corresponding continuous range is 31.7%–47.7%.

Among the climate-action personas, **climate_justice_youth** is highest on both instruments (continuous 0.2454, dual 1.3072); it is also the only non-conspiracy H persona with any observed k^* , a 2.2% dual rate (one of 46 live cells). **mitigation_first_policy** is lowest in that family (continuous 0.0464, dual 0.7571). Within science/environment, **biodiversity_food_security** is highest (continuous 0.2115, dual 1.1470), while **climate_scientist** has the lowest continuous MPR (0.0130) and **science_journalist** the lowest dual MPR (0.6745). These differences are visible by article in Figures 10 and 12 and by topology in Figure 15. Figure 11 is also important methodologically: maximum MI can be high when average persistence is modest, so a transient peak should not be reported as a sustained firestorm.

5.2.2 Heterogeneous compositions

Figures 13 and 14 provide the corresponding mix-by-article heatmaps for mean MPR and maximum MI, and Figure 16 crosses each mix with topology. The conspiracy content of the mix tracks the principal MPR gradient more closely than the generic He label. The most direct contrast is between **mix_00** (four conspiracy plus four climate/environment personas) and **mix_02** (zero conspiracy plus eight science/climate personas). **mix_00** reaches continuous MPR 1.7712 and dual-discrete MPR 3.1321, compared with 0.2645 and 1.1882 for **mix_02**. Its k^* rates are 12.8% continuous and 57.5% dual, whereas **mix_02** is 0.0% on both.

Table 5.6: Article-level live-cell MPR by arm and instrument.

Article	H continuous	H dual	He continuous	He dual
scopex_2017	0.7322	1.5390	0.7694	2.3718
chemtrails_gates_2018_2021	1.3541	2.1383	1.7852	2.3885
sai_geoengineering	0.1974	1.3576	0.1523	1.1160
paris_agreement	1.3357	1.7321	1.6821	2.4591
climate_consensus	1.5934	2.2610	1.8896	2.8991
polar_bears	0.0185	1.0399	0.0415	1.1444

`mix_01`, with two conspiracy personas, remains high (continuous 1.4770, dual 2.7795) and has the highest dual k^* rate, 58.5%. A single conspiracy persona in `mix_03` is associated with MPR 0.8945 continuous and 2.0410 dual, with k^* rates 10.3% and 33.3%. The two conspiracy-adjacent compositions do not produce a monotone dose response: `mix_04` is lower than `mix_05` continuously (0.7021 versus 1.0134), but slightly higher in dual-discrete scoring (1.6896 versus 1.6713). Nor is `mix_00` uniformly above `mix_01`: its MPR is higher, but its dual k^* rate is lower by 1.0 percentage point. The defensible result is therefore compositional—conspiracy-heavy mixes generally score above the conspiracy-free mix—rather than a universal claim that heterogeneous networks amplify or suppress a cascade.

5.2.3 Article valence and topology

Article effects cut across persona composition. Figures 21 and 22 collapse the six articles separately for H and He, while Figure 17 shows the topology-by-article pattern. The exact article means are:

`climate_consensus` is highest in all four arm-instrument combinations: H continuous 1.5934, H dual 2.2610, He continuous 1.8896, and He dual 2.8991. `polar_bears` is lowest in all four (0.0185, 1.0399, 0.0415, and 1.1444, respectively). The chemtrails-Gates article is consistently above SCoPEX: 1.3541 versus 0.7322 in H continuous, 2.1383 versus 1.5390 in H dual, 1.7852 versus 0.7694 in He continuous, and 2.3885 versus 2.3718 in He dual. However, “valence” bundles topic, framing, and fit with a persona’s prior narrative. Figures 10 and 13 show that it is an interaction, not an article-only treatment.

5.2.4 k^* and dead-cell sensitivity

Figure 19 summarizes irreversible crossings over eight ticks. Collapsed H k^* rates are 14.2% continuous and 20.1% dual; collapsed He rates are 7.6% and 32.1%. The reversed continuous/dual H-He ordering is further evidence against assigning a single causal meaning to the arm label. It also shows why k^* should not be inferred from MPR alone: the dual He arm has both a much higher crossing rate and a different scoring scale. With only eight ticks, “no k^* ” means no observed irreversible crossing within the window, not proof that a later crossing is impossible.

Figure 18 reports the missingness generated after real LLM calls. Of 1,728 expected thesis cells, 292 are dead ($\text{nScored} \leq 1$); they remain hatched and are never converted to zeros. By arm and instrument, dead rates are 16.7% for H continuous (480/576 live), 17.4% for He continuous (238/288 live), 17.0% for H dual (478/576 live), and 16.7% for He dual (240/288

live). Missingness is not uniform by topology. The largest rate is polarised He continuous at 30.6% (25/36 live), followed by hierarchical H continuous and hierarchical He continuous, both 27.8% (52/72 and 26/36 live), and polarised He dual at 27.8% (26/36 live). At the low end, linear-chain H dual is 5.6% dead (68/72 live), and ring He is 8.3% dead in both instruments (33/36 live).

Accordingly, the means in Figures 10–17 and 21–22 describe surviving scored cascades. The dead-rate pattern is part of the result: a pale heatmap cell means low MPR among live runs, while a hatched cell supplies no MPR estimate. Neither dead He cells nor the 0.0% k^* rate of `mix_02` demonstrates that heterogeneous or expert-inclusive networks are immune. Together, Figures 10–22 support a narrower conclusion: persona and article composition strongly structure observed persistence, the dual instrument usually scores higher and crosses k^* more often, and topology-dependent dead cells limit causal interpretation of the H/He contrast.

5.3 Topology results

This section reports the eight-topology sweep and comparison C1, which holds the scoring mode fixed while comparing homogeneous (H) and heterogeneous (He) identity assignments. The harvest contains one graph seed (`graphRandomSeed=42`; $N = 1$), so all differences below are descriptive properties of these runs. They do not identify a causal effect of topology or identity composition.

There were 72 candidate H cells per topology and mode (12 personas $\times$ 6 articles) and 36 candidate He cells (6 mixes $\times$ 6 articles). A cell was classified as dead when `nScored` ≤ 1 after the LLM calls. Dead cells were retained and hatched in the grids, but were neither replaced by zero nor included in the means. Thus, the denominators in Tables 5.7 and 5.8 vary by topology. Across the full 1,728-cell harvest, 292 cells were dead (rather than the preliminary brief’s 290).

5.3.1 Continuous instrument

Table 5.7 reports the continuous-only (T2c) headline separately from the dual-discrete results. Among H cells, the topology means ranged from 0.7215 for small world (56/72 live) to 0.9205 for scale-free (60/72 live). Among He cells, they ranged from 0.6593 for random ER (29/36 live) to 1.5242 for hierarchical (26/36 live). These extrema describe live cells only; for example, hierarchical He had 10 dead cells and polarised He had 11.

The continuous C1 sign was not uniform. He exceeded H on four of eight topologies (small world, echo chamber, polarised, and hierarchical), whereas H exceeded He on linear chain, ring, random ER, and scale-free. Giving each topology equal weight, the mean $\Delta(\text{He} - \text{H})$ was 0.1725. Pooling live cells instead gave H 0.8643 ($n = 480/576$ live) and He 1.0192 ($n = 238/288$ live), a difference of 0.1549. The two summaries differ because live-cell counts vary; neither treats the 146 dead continuous cells as zeros.

The near-balanced signs are visible in the continuous panel of Figures 5.1 and 5.2. An exploratory signed-rank summary of the eight topology means was $W = 14$, $p = 0.6406$, with

5 Results

Table 5.7: Continuous (T2c) live-cell mean **meanMI** by topology and C1 same-mode difference. “Live” gives the scored numerator and full-cell denominator; dead is the excluded-from-mean count. $\Delta = \text{He} - \text{H}$. $N = 1$.

Topology	H live	H dead	H mean	He live	He dead	He mean	Δ
linear chain	62/72	10	0.9129	32/36	4	0.8638	-0.0491
ring	65/72	7	0.9160	33/36	3	0.8760	-0.0400
random ER	57/72	15	0.8736	29/36	7	0.6593	-0.2143
small world	56/72	16	0.7215	31/36	5	0.7545	0.0330
scale-free	60/72	12	0.9205	30/36	6	0.8157	-0.1048
echo chamber	63/72	9	0.8490	32/36	4	1.3775	0.5284
polarised	65/72	7	0.8356	25/36	11	1.4137	0.5780
hierarchical	52/72	20	0.8751	26/36	10	1.5242	0.6490

Table 5.8: Dual-discrete (T2d) live-cell mean **meanMI** by topology and C1 same-mode difference. “Live” gives the scored numerator and full-cell denominator; dead is the excluded-from-mean count. $\Delta = \text{He} - \text{H}$. $N = 1$.

Topology	H live	H dead	H mean	He live	He dead	He mean	Δ
linear chain	68/72	4	1.6127	33/36	3	2.0852	0.4725
ring	61/72	11	1.6402	33/36	3	2.1768	0.5366
random ER	56/72	16	1.7809	29/36	7	2.0412	0.2602
small world	59/72	13	1.5977	32/36	4	1.9096	0.3120
scale-free	57/72	15	1.7034	29/36	7	1.9924	0.2890
echo chamber	61/72	11	1.7984	30/36	6	1.9632	0.1647
polarised	57/72	15	1.7005	26/36	10	1.8830	0.1825
hierarchical	59/72	13	1.6315	28/36	8	2.5934	0.9618

median delta -0.0035 ($n_+ = 4$, $n_- = 4$). These eight means come from one seed and are not independent experimental replicates; the calculation is therefore a compact description, not replicate-level inference.

5.3.2 Dual-discrete instrument

Table 5.8 reports the discrete headline from dual runs (T2d). It is a different instrument from T2c and is not averaged with the continuous headline. H means ranged from 1.5977 for small world (59/72 live) to 1.7984 for echo chamber (61/72 live). He means ranged from 1.8830 for polarised (26/36 live) to 2.5934 for hierarchical (28/36 live).

He exceeded H on all eight dual-discrete topology means. The topology-equal mean difference was 0.3974. Pooling live cells gave H 1.6814 ($n = 478/576$ live) and He 2.0800 ($n = 240/288$ live), a difference of 0.3986; the 146 dead dual cells were not imputed. The largest topology-level gap was hierarchical (0.9618), and the smallest was echo chamber (0.1647).

For completeness, the exploratory signed-rank summary across the eight topology means was $W = 0$, $p = 0.007812$, with median delta 0.3005 ($n_+ = 8$, $n_- = 0$). Because $N = 1$, this is not evidence of a general identity-mix effect. In particular, collapsed H combines conspiracy personas with low-scoring scientist personas, while collapsed He combines conspiracy-heavy and conspiracy-free mixes. The observed ordering therefore does not support a causal claim that heterogeneity increases or buffers firestorms.

The article-resolved topology pattern is shown in `fig17_topology_article_mpr.png`; topology-by-persona and topology-by-mix views are in `fig15_persona_topology_mpr.png` and `fig16_mix_topology_mpr.png`. The corresponding denominators and hatch pattern are

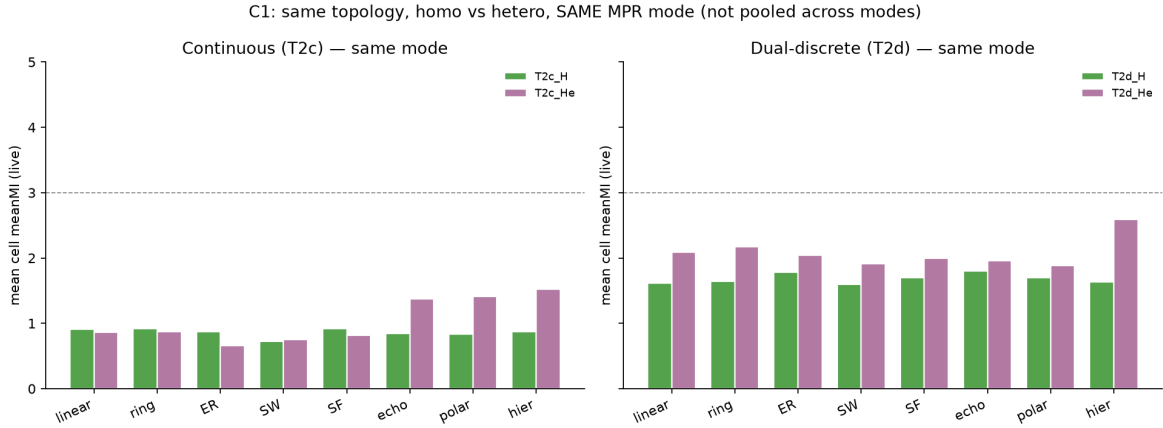


Figure 5.1: C1 same-mode H and He live-cell means for all eight topologies. Continuous and dual-discrete panels remain separate; dead cells are excluded from means rather than set to zero. $N = 1$.

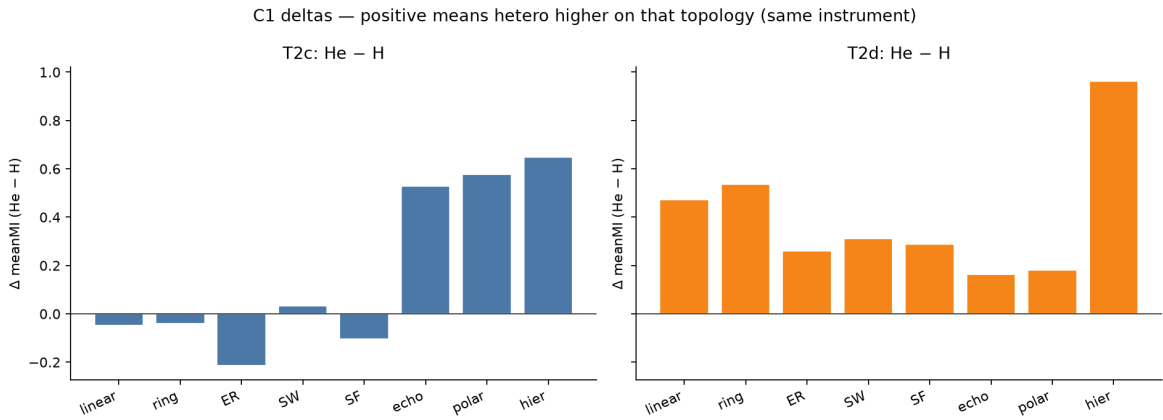


Figure 5.2: C1 topology-level $\Delta(\text{He} - \text{H})$. Continuous has four positive and four negative differences; dual-discrete has eight positive differences. These are descriptive one-seed contrasts.

visualised in `fig18_dead_rates.png`. Those figures should be read with the same live-only and mode-separation rules used in the tables above.

5.4 Results on the 63-node D-net

5.4.1 Graph and persona placement

The D-net experiments used the custom graph in `thesisExperiment/data/derived/debnath_hashtag_cascade.json`. It contains 63 nodes and 228 directed edges, collapsing to 171 unique undirected pairs. The nodes are accounts keyed by published hashtags and the edges encode hashtag co-occurrence and same-cluster links. Although the edges occupy the importer’s `retweets[]` field, they are not observed retweets: Twitter/X hydration failed, and the object is not a retweet cascade. Both D-net persona conditions used exactly this topology and the same seed node, `user_chemtrails_hub`.

The heterogeneous placement retained the reconstructed Debnath type mix. Of the 63 placements, 56 used the exact reconstructed persona identifier. The remaining seven used a coarser persona identifier while retaining the mapped Debnath type:

- `user_depopulation_amp`: `conspiracy_depopulation` $\rightarrow$ `conspiracy_believer`;
- `user_climateaction_piggyback_amp` and `user_climateaction_piggyback_peri_1`: `conspiracy_climate_piggyback` $\rightarrow$ `conspiracy_believer`;
- `user_biodiversity_hub`, `user_biodiversity_peri_1`, and `user_foodsecurity_amp`: `biodiversity_food_security` $\rightarrow$ `environmental_concern`; and
- `user_mitigation_amp`: `mitigation_first_policy` $\rightarrow$ `climate_action_advocate`.

Thus the heterogeneous arm placed 28 conspiracy-family, 19 climate-action, 13 environmental, and three expert personas. The homogeneous arm instead assigned `conspiracy_believer` to all 63 nodes. The empirical hashtag identities were 28 conspiracy, 19 climate-action, 14 environmental-concern, and two other nodes; this difference in counting reflects the coarser simulation placement, including the type-only fallback for `user_mitigation_amp`, rather than a change to the graph.

5.4.2 Four D-net arms

Four completed configurations crossed continuous versus dual-discrete auditing with homogeneous conspiracy versus heterogeneous Debnath-mapped personas. Each configuration propagated two seed articles, producing the eight article-level results in Table 5.9. These are auditor scores from the simulation, not measurements made on Twitter.

Table 5.9: Article-level auditor scores in the four D-net configurations. The continuous sidecar from a dual run is shown only as a diagnostic and is not interchangeable with the continuous-run headline.

Configuration	Article	Headline score	<i>n</i> scored	Dual sidecar
Dnet_c_H_conspiracy	SCoPEX	1.7349	970	—
Dnet_c_H_conspiracy	chemtrails-Gates	3.3378	984	—
Dnet_c_He_mixed	SCoPEX	0.9983	591	—
Dnet_c_He_mixed	chemtrails-Gates	2.5068	1,035	—
Dnet_d_H_conspiracy	SCoPEX	4.2537	1,076	2.2495
Dnet_d_H_conspiracy	chemtrails-Gates	3.7303	749	3.1251
Dnet_d_He_mixed	SCoPEX	2.6880	407	1.1302
Dnet_d_He_mixed	chemtrails-Gates	1.5297	1,027	1.1497

Within each article and instrument, homogeneous conspiracy scored above the mixed placement: 1.7349 versus 0.9983 and 3.3378 versus 2.5068 under the continuous auditor, and 4.2537 versus 2.6880 and 3.7303 versus 1.5297 under the dual-discrete auditor. Article ordering depended on the instrument. Chemtrails-Gates exceeded SCoPEX in both continuous arms, whereas SCoPEX exceeded chemtrails-Gates in both dual-discrete arms.

5.4.3 Do the small-topology findings hold?

Only the composition result transfers cleanly. In the eight-node grid, conspiracy-family homogeneous cells were the high-score family on both instruments (3.1691 dual-discrete and 2.2194 continuous). The D-net result is consistent with that finding: replacing the mixed

type placement with 63 conspiracy-believer seats raised the score for both articles under both auditors.

The collapsed H/He result does not transfer as an H/He law. Across the eight small topologies, collapsed He exceeded collapsed H by 0.1549 under continuous scoring and by 0.3986 under dual-discrete scoring, although the continuous He>H ordering held on only four of the eight individual topologies. On D-net, H exceeded He in all four like-for-like article and instrument comparisons. This reversal is consistent with the different composition of the labels: small-topology H pooled conspiracy personas with scientists, while D-net H means conspiracy-only.

Nor does the small-grid instrument ordering hold without exception. The eight-topology aggregates placed dual-discrete above continuous on every topology. On D-net, the dual headline exceeded the corresponding continuous-run headline for homogeneous SCoPEX, homogeneous chemtrails–Gates, and mixed SCoPEX, but not for mixed chemtrails–Gates (1.5297 dual-discrete versus 2.5068 continuous). Because these are different auditors, even the three concordant comparisons are instrument differences rather than estimates in a shared unit. D-net has only one fixed 63-node topology, so it cannot replicate or rank the eight small topology treatments.

5.4.4 Auditor MI is not empirical Twitter MPR

Debnath provides no empirical 0–5 misinformation-propagation-rate (MPR) outcome for this graph. The source tweet identifiers were not hydrated, the graph has no empirical tweet clock, and no tweet-level content score can be paired with a simulated event. Accordingly, the values in Table 5.9 are LLM-as-judge auditor misinformation indices (MI) inside the simulation. They must not be described as Twitter MPR, as Twitter MI, or as a match to observed misinformation. The dual-discrete headline, its continuous sidecar, and the continuous-only headline are also separate instruments and are not averaged.

5.4.5 Structural comparison and validation diagnostics

The copied graph has global transitivity 0.3146, mean local clustering 0.5153, and mean unique degree 5.4286. The `user_chemtrails_hub` node has 33 unique neighbours. The conspiracy cut has modularity 0.4039 under mixed labels and 0.4566 when all nodes are labelled conspiracy. Mixed-label identity homophily is 0.8377 in the simulation; homogeneous identity homophily is 1.0 by construction, so the latter is not evidence of an emergent echo chamber.

For the rooted cascade-shape comparison, the empirical object had depth 4, breadth 33, reached size 61, and structural virality 2.7. Across 16 simulated D-net cascades, the corresponding means were 2.5625, 40.9375, 53.5, and 1.7963. Breadth was the only one of the three validation metrics (depth, breadth, and structural virality) within the pre-specified 30% band; the resulting structural-similarity score was 0.6885.

The distributional diagnostics do not validate a digital twin. With one empirical object and 16 simulated cascades, the Kolmogorov–Smirnov results were $D = 1.0000$, $p = 0.1104$ for depth; $D = 0.8125$, $p = 0.2954$ for breadth; and $D = 1.0000$, $p = 0.1104$ for structural virality.

Jensen–Shannon divergence was 1.0000 for all three metrics, and all three distribution-match flags were false. The empirical sample size of one makes the KS comparisons underpowered; their non-significant p -values are not evidence of equivalence. Finally, DTFS was 0.2754, below the 0.70 validation threshold (`isValidated=false`). Its content-correlation component was fixed at zero because no empirical MPR or tweet-text MI exists. The structural evidence therefore supports a comparison of cascade shape, not empirical misinformation fidelity or digital-twin validation.

5.5 Applying the Pfeffer firestorm framework

5.5.1 The original framework and the thesis remapping

Pfeffer, Zorbach, and Carley (2014) define an online firestorm as a sudden discharge of large quantities of negative word-of-mouth and complaint messages directed at a person, company, or group. The messages are predominantly opinion rather than fact and are therefore highly affective. The article is a qualitative–synthetic account based on observed cases; it does not present an agent-based model, a tipping-point estimator, or a climate-discourse experiment.

The Outlook of Pfeffer, Zorbach, and Carley (2014) identifies seven interrelated factors:

1. *speed and volume of communication*;
2. *binary choices*;
3. *network clusters*;
4. *unrestrained information flow*;
5. *lack of diversity*;
6. *cross-media dynamics*; and
7. *network-triggered decision processes*.

These are the original factors used here. In particular, valence and surprise are not numbered Outlook factors. Valence is motivated by the affective character in the paper’s definition of a firestorm, while surprise is a thesis operationalisation of the novelty, speed, and volume mechanism in factor 1.

The thesis vocabulary—valence, surprise, identity, clustering, echo, temporal, and cross-media—is consequently a computational remapping, not a renaming of Pfeffer et al.’s list. The first six terms describe variables or observables that can at least partly be represented in the simulation; cross-media is included as a seventh ledger row but is held because the model has no legacy-media layer. Conversely, Pfeffer et al.’s binary-choice factor has no dedicated thesis row. The agents choose among **forward**, **reinterpret**, and **drop**, so the engine is ternary rather than an implementation of the paper’s like/share/retweet either–or mechanism.

Table 5.10: Original Outlook factors and their treatment in the thesis. The right-hand column is an operational mapping, not Pfeffer et al.’s terminology.

#	Pfeffer et al. (2014)	Thesis treatment
1	Speed and volume of communication	Surprise (shock versus drip) and temporal horizon; surprise is held at drip and empirical timing is unavailable.
2	Binary choices	No dedicated observable or sweep; simulated actions are ternary.
3	Network clusters	Clustering and part of the echo measurement.
4	Unrestrained information flow	Partly represented by echo and hub reach, but not the hundreds or thousands of weak ties discussed in the paper.
5	Lack of diversity	Identity composition and part of echo/homophily; homogeneous versus mixed belief profiles are compared.
6	Cross-media dynamics	Held; neither the empirical reconstruction nor the engine contains a legacy-media or newsroom layer.
7	Network-triggered decision processes	Related to the temporal ledger row, but the four adoption stages are not logged or estimated.

5.5.2 Order of application: empirical structure before simulation

The application proceeds in two distinct stages. First, structural observables are calculated on the empirical fallback derived from Debnath et al. (2023b). This object is a hashtag co-occurrence graph with 63 nodes, 228 directed edges, and 171 unique undirected pairs. It is not a retweet, reply, follower, or `conversation_id` cascade. The 814,924 tweet identifiers in the associated dataset were not hydrated, and the graph contains no usable tweet clock. Its labels and edges can therefore support a bounded comparison of hashtag structure, but not a reconstruction of user-to-user diffusion.

Second, D-net imports the same 63-node, 228-edge topology and runs agents on it. This ordering matters: transitivity, local clustering, hub degree, and the mixed-label conspiracy cut already exist before any simulated message is generated. D-net copies this structure; it does not produce or validate it. The post-simulation evidence consists of content-auditor scores, realised actions and reach on that fixed graph, together with label-dependent quantities for the homogeneous and mixed assignments.

Empirical hashtag structure. The graph has global transitivity 0.315, mean local clustering 0.515, and mean unique degree 5.43. The `#chemtrails` hub has 33 unique neighbours. With the empirical discourse labels, the conspiracy-cut modularity is $Q = 0.404$. Identity edge homophily is 0.873; belief-profile-family homophily is 0.838. The identity composition is 28 conspiracy, 19 climate-action, 14 environmental, and two expert/other nodes. Thus conspiracy discourse is a plurality ($28/63 = 0.444$), not a monoculture.

D-net after simulation. The mixed D-net condition retains 28 conspiracy nodes and the empirical type-level composition, with 19 climate-action, 13 environmental, and three expert nodes after belief-profile assignment. The homogeneous condition instead assigns all 63 seats the `conspiracy_believer` profile. Because both conditions use the same edge set, transitivity, local clustering, and hub degree remain 0.315, 0.515, and 33. Mixed D-net identity homophily

Table 5.11: Interpretive boundary of the Pfeffer application. Empirical values refer to the hashtag graph before simulation; simulated values refer to D-net after a run.

Observable	What the method reveals	What the method cannot reveal
Clustering	Before simulation, the hashtag construction contains substantial closure: transitivity 0.315, local clustering 0.515, and a hub of degree 33. The mixed-label cut has $Q = 0.404$.	These values are not retweet triadic closure or emergent D-net clusters. The graph is a documented co-occurrence construction, not a Watts–Strogatz fit, and previously reported Debnath eigenvector centralities were not re-estimated here.
Echo	Empirical identity homophily 0.873 and belief-profile-family homophily 0.838 show that most represented ties remain within discourse blocks. Mixed D-net retains comparable block echo (0.838).	The measure does not identify algorithmic ranking, weak-tie exposure, or an emergent chamber. Homogeneous D-net homophily of 1.0 is tautological and cannot measure echo formation.
Identity	The comparison separates a mixed empirical-type composition (conspiracy share 0.444) from the homogeneous intervention (share 1.0). It therefore isolates identity composition as a model input on a common topology.	The labels are theory-grounded reductions, not HDBSCAN communities recovered from hydrated tweets, and they do not establish individual users' identities or follower-network homophily.
Valence proxy	The empirical side supplies only the paper-reported toxicity mean (approximately 0.17) and a hashtag prior proxy (0.1097); the simulated side supplies an LLM-auditor misinformation index. On the continuous instrument, the chemtrails seed has higher mean MI than the SCoPEX seed in both homogeneous and mixed D-net.	Toxicity, a hashtag prior, and auditor MI are different constructs and scales. Debnath provides no empirical MPR. The comparison cannot be described as Twitter MI, an empirical-to-simulated MPR match, or validation of a firestorm.
Surprise	The ledger records the seeded regime: a single <code>user_chemtrails_hub</code> provides drip seeding. The paper-reported SCoPEX volume spike motivates the construct.	Surprise is held, not tested. There is no all-node shock condition and no reconstructed empirical volume series, novelty half-life, or attention decay.
Temporal	D-net records engine progression for eight ticks and shows 61/63-node reach within that budget.	Empirical timing is unavailable because the hashtag fallback has no tweet clock. Ticks cannot be translated into minutes or hours, and the method cannot estimate inter-arrival times, half-life, or Pfeffer et al.'s knowledge–persuasion–propagation–affirmation sequence.
Cross-media	The ledger makes the omission explicit rather than attributing it to a topology.	Cross-media dynamics are held on both sides. There is no social-to-legacy- media-to-social loop, newsroom agent, external-volume estimate, or basis for treating a hierarchical graph as cross-media amplification.

is 0.838, whereas the homogeneous value is 1.0 by construction. The latter is not evidence that an echo chamber emerged. It is the mechanical result of giving every node the same identity. The corresponding conspiracy-cut modularities are $Q = 0.404$ for mixed labels and $Q = 0.457$ under homogeneous labels.

Across the four primary D-net cells, 61 of 63 nodes receive a message within the held eight-tick horizon. This demonstrates reach under the engine's rules on the imported graph. It does not establish that the empirical Twitter discourse spread to 61 users in eight equivalent units of time. Simulation ticks have no defensible conversion to hours, and the eight-hop budget is a computationally compressed horizon.

5.5.3 What the application can and cannot reveal

The valence result requires particular care. In the continuous runs, mean auditor MI for SCoPEX is 1.735 in homogeneous D-net and 0.998 in mixed D-net; for the chemtrails seed it is 3.338 and 2.507, respectively. These values show content degradation under a particular auditor and graph intervention. They do not measure indignation or toxicity. The dual-discrete

instrument is a separate scale and does not preserve the same article ordering in every cell, so the two instruments are not averaged.

Likewise, the network-mean threshold k^* , where used, is a thesis rule based on simulated MI. It is neither a statistic introduced by Pfeffer, Zorbach, and Carley (2014) nor an empirical estimate from Debnath et al. (2023b). Structural cascade comparison also does not repair this gap: the available comparison has one empirical graph, and its reported digital-twin fit score is 0.2754, below the 0.70 validation threshold. The D-net application should therefore be read as a controlled, theory-mapped simulation on an empirical structural stand-in, not as a validated digital twin of the geoengineering conversation.

5.5.4 Interpretation

The application supports three bounded conclusions. First, clustering and block homophily are properties of the empirical hashtag representation before simulation; they are consistent with the structural mechanisms in Pfeffer et al.’s factors 3 and 5, but do not prove a user-level firestorm. Second, holding topology fixed while changing the belief-profile composition distinguishes mixed from homogeneous D-net outcomes and avoids mistaking an all-identical assignment for emergent echo. Third, content auditing can describe simulated factual degradation after propagation, but it cannot supply the missing empirical temporal, binary-choice, weak-tie, or cross-media processes. The framework is thus applied as an explicit mapping and limitation ledger, not as a claim that all seven original factors were experimentally realised.

6 Discussion

This chapter interprets the Phase 2 results in relation to the research questions and the literature. The distinction between evidence, interpretation, and implication is maintained throughout. This distinction is necessary because the campaign comprises one run per configuration ($N = 1$), one graph seed, an eight-tick horizon, and 292 dead cells among 1,728 configured cells. The reported quantities are therefore descriptive properties of the present LLM simulation. They are neither estimates of population effects nor measurements of misinformation on Twitter.

6.1 Synthesis of the research questions

6.1.1 RQ1: What does the instrument comparison establish?

The measurement result precedes all substantive comparisons. Dual-discrete scores exceeded continuous scores on every topology in both arms, while topology ranks aligned weakly. The campaign therefore shows that MI and MPR conclusions depend on the auditor regime. This finding answers RQ1: neither instrument can serve as an unqualified ground truth, and cross-mode pooling is not admissible.

6.1.2 RQ3: How does network topology affect misinformation propagation?

Evidence. Topology changed the live-cell scores, the frequency of irreversible k^* , and the probability that a cascade remained scorable. However, no topology occupied the highest or lowest position consistently across identity composition and scoring instrument. In homogeneous continuous runs, scale-free networks had the highest topology mean (0.9205), whereas small-world networks had the lowest (0.7215). In homogeneous dual-discrete runs, echo chambers had the highest mean (1.7984), again with small-world networks lowest (1.5977). In heterogeneous runs, the hierarchical topology had the highest mean under both instruments (1.5242 continuous and 2.5934 dual-discrete). These latter estimates must be read together with the hierarchical dead-cell rates of 27.8% and 22.2%, respectively. Polarised heterogeneous continuous runs likewise combined a high live-cell mean (1.4137) with a 30.6% dead-cell rate.

Interpretation. The results do not support a topology-invariant ranking. Topology appears to condition which messages are observed and scored, but it does so jointly with persona placement, article content, and cascade survival. In particular, live-only means on topologies with many dead cells describe the surviving cascades rather than the generator as a whole. The scale-free versus random comparison is primarily a hub and degree-sequence contrast; it is not a controlled comparison of clustering coefficients. Similarly, the hierarchical generator represents

authority-oriented transmission and should not be interpreted as a model of cross-media dynamics.

Implication. Topology should be treated as a conditioning structure, not as a sufficient cause of factual degradation. Subsequent work should use repeated graph seeds, report cascade survival as an outcome, and match graph properties when isolating a specific mechanism such as clustering, centralisation, or community separation. On the present evidence, selecting a single “most harmful” topology would overstate what the design identifies.

6.1.3 RQ2: What is the role of identity composition?

Evidence. When all eight generated topologies were collapsed within one instrument, the heterogeneous arm exceeded the homogeneous arm: by 0.1549 in continuous MI (H 0.8643, He 1.0192) and by 0.3986 in dual-discrete MI (H 1.6814, He 2.0800). The continuous direction was nevertheless split: He exceeded H on four of eight topologies, whereas H exceeded He on the other four. The dual-discrete direction was He>H on all eight topologies. Composition-specific results were substantially more differentiated than these arm labels. Homogeneous conspiracy-family cells reached 2.2194 continuous and 3.1691 dual-discrete, compared with 0.1118 and 0.8687 for homogeneous science/environment personas. Among heterogeneous mixes, `mix_00`, containing four conspiracy personas, scored 1.7712 continuous and 3.1321 dual-discrete; conspiracy-free `mix_02` scored 0.2645 and 1.1882.

Interpretation. The collapsed H/He contrast is not an estimate of a general effect of diversity. The homogeneous arm contains both conspiracy-only and scientist-only societies, while the heterogeneous arm contains mixes with different conspiracy shares and different persona identities. Averaging within these broad labels changes the distribution of substantive roles. Consequently, the observed He>H collapse is compatible with a strong composition gradient: conspiracy-conditioned cells score high, whereas science-conditioned cells score low. It does not establish that heterogeneity intensifies misinformation, just as the low value of `mix_02` does not establish that diversity prevents it.

Implication. Future comparisons should hold the proportion and identity of conspiracy-conditioned seats constant while varying only within-network diversity or placement. Network means should be accompanied by persona-conditioned MPR and exposure-matched comparisons. This would separate a genuine interaction between heterogeneous neighbours from the mechanical dilution or enrichment produced by changing the number of high-scoring personas.

6.1.4 Measurement implications across RQ2 and RQ3

Evidence. Dual-discrete scores exceeded continuous scores on all eight topologies in both arms. The topology-mean difference was 0.8201 for homogeneous and 1.0450 for heterogeneous runs. Among cells live under both modes, the median paired difference was 0.8125 for H ($n = 398$) and 0.9778 for He ($n = 198$). Instrument choice also affected the incidence of irreversible k^* : the collapsed He rates were 32.1% under dual-discrete scoring and 7.6% under

continuous scoring. Moreover, topology rankings agreed only weakly across modes in the exploratory analysis (Spearman $\rho = 0.095$ for H and $\rho = 0.167$ for He).

Interpretation. The two modes are not interchangeable measurements on a common numerical scale. The discrete instrument awards categorical penalties for omitted or contradicted answers, whereas the continuous instrument resolves degradation differently. The systematic level difference and weak rank agreement show that instrument choice can affect both absolute severity and comparative claims about topology. The continuous sidecar generated during a dual run is useful for diagnosing disagreement, but it is not a third independent MPR.

Implication. Substantive conclusions should be reported separately for each instrument. Averaging the two modes would create a quantity without a defined auditor. The disagreement also motivates external validation: a preregistered sample of rewrites should be assessed by human coders, and calibration should examine whether each instrument preserves article- and persona-level ordering rather than merely correlating at the event level. Until such validation is available, k^* is instrument-relative.

6.1.5 RQ4: Does the D-net analysis generalise the generated-topology findings?

Evidence. D-net used a 63-node custom graph reconstructed from hashtag co-occurrence, not a retweet cascade. Its homogeneous condition assigned `conspiracy_believer` to all 63 seats; its mixed condition retained a conspiracy share of 28/63. On the identical edge set, homogeneous conspiracy exceeded the mixed condition for both articles and both instruments: continuous SCoPEx 1.7349 versus 0.9983, continuous chemtrails–Gates 3.3378 versus 2.5068, dual-discrete SCoPEx 4.2537 versus 2.6880, and dual-discrete chemtrails–Gates 3.7303 versus 1.5297. Structural comparison was less favourable. Breadth fell within the specified 30% band, whereas depth and structural virality did not; DTFS was 0.2754 and failed the 0.70 validation threshold. Tests comparing one empirical graph with 16 simulated cascades were underpowered.

Interpretation. D-net provides a useful stress test of the composition interpretation because the topology is held fixed while persona labels change. It does not provide out-of-sample validation of the eight-node generators, and it is not a digital twin of Debnath et al.’s Twitter corpus. The graph imports co-occurrence structure and then simulates message propagation on that structure; it neither reconstructs empirical retweet paths nor reproduces an empirical misinformation score. The D-net result therefore supports the narrow statement that a conspiracy-only assignment scored above the retained mixed assignment on this larger custom graph.

Implication. The qualitative composition gradient travels from the generated graphs to D-net, but numerical effect sizes and cascade-shape claims do not. Stronger generalisation would require hydrated interaction data, common community and centrality estimators, temporal

information, multiple reconstructed cascades, and repeated simulation seeds. Simulated auditor MI must remain separate from Debnath et al.’s toxicity and network measures (Debnath et al., 2023b; Ruths and Pfeffer, 2014).

6.2 Reconciling the collapsed and D-net orderings

Evidence. The generated-topology collapse gives $H_e > H$. In D-net, homogeneous conspiracy scores exceed the mixed assignment. These comparisons use different definitions of H . Across the eight-node grid, H denotes twelve separate homogeneous persona societies, including four conspiracy personas as well as climate-action, environmental, journalist, and scientist personas. D-net H denotes one specific treatment: all 63 nodes are assigned the `conspiracy_believer` persona. D-net H_e preserves a majority non-conspiracy composition. Restricting the generated-topology H cells to the conspiracy family already reverses the broad collapse: conspiracy H is well above overall H_e on both instruments.

Interpretation. There is therefore no empirical contradiction to resolve. The label H encodes within-network uniformity, but it does not encode which identity is uniform. Collapsing across homogeneous scientist and homogeneous conspiracy societies answers a different question from comparing a conspiracy monoculture with a mixed society. D-net changes both diversity and the prevalence of the high-scoring persona when moving from H to H_e . Its ordering is consequently consistent with composition, but cannot distinguish composition from a causal benefit of heterogeneity.

Implication. H and H_e should not be used as portable substantive treatments without an attached composition descriptor. A defensible notation would distinguish, for example, H -conspiracy, H -science, H_e -44%-conspiracy, and H_e -0%-conspiracy. A matched factorial design could then vary diversity while holding conspiracy share, seed position, and topology fixed. This reformulation is more informative than asking whether H or H_e is “higher” in the abstract.

6.3 Relation to the award paper and adjacent literature

6.3.1 The CIKM/LASS study as prior method, not a benchmark result

Evidence from the literature. Maurya et al. (2025) introduced the auditor–node framework, MI and MPR, and the severity taxonomy used by the present system. Its published evaluation concerned 30-hop linear branches, a different article collection, a larger persona inventory, ten auditor questions, and a different model configuration. It reported that identity- and ideology-oriented homogeneous branches could accelerate degradation and that experts could stabilise it; it also reported propaganda-tier outcomes in approximately 85% of heterogeneous branch–domain pairs. Thus, the award paper itself does not establish a universal rule that heterogeneous societies must score below homogeneous societies.

Interpretation of the present comparison. The present study transfers the instrument to climate and geoengineering content and extends the setting from linear branches to generated and custom graphs. The non-zero, persona-sensitive scores indicate that the instrument remains responsive in this domain. The observed collapsed He>H ordering is not a failed replication of a universal CIKM H>He hypothesis, because no such universal hypothesis follows from the prior heterogeneous result. Conversely, D-net H-conspiracy>He-mixed cannot be presented as a replication: identity inventories, graph objects, hop budgets, auditor questions, and model settings differ, and the present campaign has only one replicate.

Implication. The defensible contribution is a boundary analysis of the prior method, not a numerical reproduction of the award paper. A direct conceptual replication would require matched linear chains, an expert-inclusive heterogeneous arm, the same climate articles in both arms, a sufficiently broad article set, and node-type-conditioned outcomes. The eight-hop graph campaign can motivate that test but cannot substitute for it.

6.3.2 Topology and diffusion literature

Evidence from the literature. Classical diffusion models establish that network structure affects reach and activation opportunities (Kempe, Kleinberg, and Tardos, 2003). Small-world and preferential-attachment models formalise different structural properties (Watts and Strogatz, 1998; Barabási and Albert, 1999). Empirical work further shows that cascade depth and breadth are not captured by size alone (Goel et al., 2016), and that false news can diffuse farther and faster than true news in observational data (Vosoughi, Roy, and Aral, 2018).

Interpretation of the present comparison. The present results are compatible with the general proposition that structure conditions diffusion, but they do not re-estimate these empirical findings. Here the state is mutable text, forwarding is mediated by an LLM, and live-cell MI is an auditor output. Topology effects are also entangled with cascade death and composition. The data therefore add a content-transformation perspective to network diffusion; they do not show that one canonical graph class is intrinsically most susceptible to misinformation.

Implication. Network controls in LLM societies should include both structural summaries and content outcomes. Matching degree, density, seed centrality, and clustering would permit more specific tests than topology names alone.

6.4 Pfeffer’s firestorm theory: insights and boundaries

Evidence. Pfeffer, Zorbach, and Carley (2014) identify seven interrelated factors: speed and volume of communication, binary choices, network clusters, unrestrained information flow, lack of diversity, cross-media dynamics, and network-triggered decision processes. The present six-knob vocabulary is a computational remapping rather than this original list. The campaign varied network structure and identity composition and compared article frames. It

held surprise as a single-node drip, fixed the horizon at eight ticks, omitted a cross-media layer, and used ternary actions—forward, reinterpret, and drop—rather than binary platform actions. D-net’s empirical structure had transitivity 0.315, mean local clustering 0.515, and mixed-label homophily close to 0.84, but these structures were pre-wired rather than generated by the simulation.

Interpretation. Two Pfeffer insights receive limited support as useful design principles. First, network clusters and identity structure cannot be reduced to a single topology label: the high-scoring composition can be concentrated or diluted within the same graph. Second, lack of diversity is not adequately tested by the binary H/He label unless identity prevalence is controlled. The D-net comparison illustrates both points particularly clearly. By contrast, this campaign does not test speed and volume as a shock manipulation, unrestrained information flow at platform scale, binary-choice dynamics, cross-media amplification, or the four stages of network-triggered adoption. Nor does k^* originate in Pfeffer’s framework; it is a thesis-specific auditor threshold.

Implication. Pfeffer’s framework is most productive here as an audit of what the simulation represents and omits. It should not be presented as a seven-factor confirmation. A fuller test would add shock versus drip seeding, a binary-action ablation, a legacy-media or exogenous-broadcast layer, and explicit logging of repeated exposure and affirmation. Such extensions would move the framework from retrospective mapping towards identified mechanism tests.

6.5 Alternative explanations and validity of the interpretation

Several alternative explanations remain plausible. Persona prompts may induce role compliance rather than persistent beliefs; high conspiracy scores may therefore partly measure how faithfully the model follows an instruction to use conspiratorial language. The action policy gives reinterpretation a substantial role, which creates opportunities for factual mutation by design. The same model family participates in generation and automated assessment, and the five author-defined questions constrain what counts as degradation. Article differences combine topic, wording, affective charge, and prior fit with persona prompts, so they are not a clean manipulation of valence. Finally, excluding dead cells from means avoids coding them as factual zeros, but produces survivor-conditioned estimates.

These limitations do not make the campaign uninformative. They narrow the appropriate claim. The experiment demonstrates that, under the documented prompts and graph protocols, auditor scores vary with persona composition, topology, and scoring mode. It does not establish human susceptibility, platform-level firestorm dynamics, or causal effects in an empirical population. Human evaluation, model ablations, repeated seeds, and a fail-closed auditor are therefore validation requirements rather than optional refinements (Zheng et al., 2023; Park et al., 2023).

6.6 Overall implications

For simulation methodology, the principal implication is that treatment labels must preserve their substantive content. “Homogeneous” and “heterogeneous” are insufficient without persona shares, placement, and node-conditioned outcomes. Likewise, topology names should be accompanied by realised graph statistics and cascade-survival rates.

For measurement, discrete and continuous MI should be treated as separate operationalisations. Agreement in a direction across both instruments is stronger descriptive evidence than a result present under only one, but neither becomes ground truth through convergence. Divergence is itself a result about the sensitivity of the conclusion to the auditor.

For empirical generalisation, D-net is an intermediate case: larger and structurally grounded in a hashtag graph, yet still neither a retweet cascade nor a temporal reconstruction. Its contribution is to show that the composition interpretation survives a change in graph size and source under the same simulator. It does not validate the simulator against Twitter.

Taken together, the research questions yield a restrained conclusion. Topology conditions propagation but has no stable winner; identity composition is more informative than the H/He label; severity and k^* are instrument-dependent; and D-net supports only qualitative, not numerical, generalisation. Pfeffer’s framework explains why these dimensions should be considered jointly, while also making visible the mechanisms that the current campaign leaves untested.

7 Validity, Ethics, and Reproducibility

The Phase 2 results are evidence about the behaviour of one configured simulation campaign, not estimates of population effects and not a validated digital twin of online climate discourse. The harvest itself records `thesisGrade=false`. Its main value is therefore diagnostic: it shows which patterns appeared under the specified graph generators, personas, articles, language model, and scoring instruments, while exposing where those patterns are sensitive to design choices.

7.0.1 Internal and statistical validity

One replicate. Every grid cell was run once with `graphRandomSeed=42` ($N = 1$). Events from the same run are dependent because they share a graph, node states, inboxes, trust updates, and article history. Article-cells and scored events consequently cannot be treated as independent experimental replicates. The reported means, rank orders, and exploratory tests are descriptive comparisons of this harvest; they do not provide confidence intervals, population-level significance tests, or causal estimates of a topology or identity-mix effect. In particular, a small p -value computed over pooled cells would not repair the absence of independent simulation replicates.

Mitigation. Results are separated by scoring mode and accompanied by live-cell counts and dead-cell rates. Claims are phrased as observations “in this harvest”, and the analysis avoids interpreting event counts as sample size. Future work should run independently seeded graph and model replicates, predefine topology-level contrasts, and estimate uncertainty with a multilevel or cluster-aware analysis whose experimental unit is the independent run.

Eight-hop horizon. The campaign stops after eight ticks/hops, a cost compression relative to the CIKM chain length of 30. This horizon is neither an empirical firestorm duration nor Debnath et al.’s protocol: their “eight” is a skip-gram context window of eight words. The short horizon right-censors propagation and weakens the interpretation of k^* , which is defined as the first observed tick above the threshold with no later observed recovery. A cascade that appears irreversible at tick eight may recover at tick nine.

Mitigation. Tick counts are not translated into clock time, and k^* is reported as an eight-tick simulation outcome. Future work should conduct a horizon sensitivity analysis (including the 30-step reference), use survival or censoring-aware summaries, and calibrate a time scale only if timestamped empirical cascades become available.

Model dependence and stochastic services. Both the agents and the auditor use `gpt-4o-mini`. The findings may therefore reflect this model’s instruction following, safety tuning, linguistic priors, and version at the time of execution. A hosted model can change

without a corresponding repository change, and API sampling may not be perfectly reproducible even when the graph seed is fixed. High misinformation scores for conspiracy personas are compatible with prompt obedience; they are not evidence that a field community would behave in the same way.

Mitigation. The model name, configuration, graph seed, raw run directories, parser rules, and harvest timestamp are retained. Future work should pin provider model snapshots where possible, record sampling and API metadata, repeat runs across several model families and temperatures, and report model-by-condition interactions rather than presenting one model as a generic human proxy.

Auditor circularity. The same model family generates the messages and judges them against five author-curated questions. Shared model priors can create correlated generator and judge errors, while the selected questions determine what counts as missing or incorrect. Discrete and continuous scores are also different instruments: the dual-run discrete headline, its continuous sidecar, and the continuous-only campaign are not interchangeable. Their disagreement shows that misinformation prevalence is partly measurement-dependent.

Mitigation. The two headlines are never averaged, dual gap is treated as an instrument diagnostic rather than a third MPR, and the question key is retained for inspection. Future validation should blind independent human coders to condition, measure inter-rater reliability, adjudicate disagreements, test alternative question sets, and use judge models from model families not used to generate the content.

Hatched-cell policy. Of 1,728 thesis cells, 292 are hatched because $nScored \leq 1$ despite real LLM calls. They are kept in the ledger but excluded from live-only means; they are neither zeros nor evidence that a mix prevented misinformation. If death depends on topology, persona, or scoring mode, complete-case means describe surviving cascades and may be selection-biased. This concern is material for slices such as hierarchical and polarised heterogeneous runs, which have comparatively high dead rates.

Mitigation. Every live mean is paired with its denominator and hatch rate, and no dead cell is imputed as zero. Future work should diagnose the mechanism producing unscored cells, predefine a minimum-information rule, rerun failed cells under the same protocol, and report sensitivity bounds or selection models rather than relying only on complete cases.

7.0.2 Construct and external validity

Descriptive comparisons, not interventions. The grid changes several consequential features at once. Homogeneous and heterogeneous labels encode different persona compositions, while generators also differ in degree sequence, reachability, and cluster assignment. Collapsed H–He differences therefore do not isolate “diversity”, and topology rank differences do not identify a causal network effect. The pooled tables concatenate cells; they do not form a physical super-graph.

Mitigation. Same-mode contrasts are kept separate, composition is shown by persona family and mix, and D-net is reported outside the eight-node pool. Future experiments should use

matched compositions, graph randomisations that preserve relevant degree sequences, factorial ablations, and preregistered estimands for identity, topology, and their interaction.

Persona reduction. The personas are theory-faithful prompt reductions of discourse types, not individuals, survey respondents, or HDBSCAN centroids reconstructed from the full Debnath corpus. Rich and changing identities are compressed into short, stable instructions. The simulation therefore risks stereotyping conspiracy-oriented, activist, scientific, or environmental participants and may overstate within-group consistency. A model’s performance of a persona must not be interpreted as a psychological diagnosis or a moral ordering of real people.

Mitigation. Results are attributed to prompt configurations rather than demographic groups, and the auditor’s scientific key evaluates claims rather than the worth of a persona. Future work should co-design personas with domain experts and affected communities, represent uncertainty and within-group variation, test multiple prompt formulations, and audit outputs for caricature and disparate treatment.

Hashtag graph, hydration failure, and absent empirical mpr. The 63-node D-net object is a hashtag co-occurrence reconstruction with 228 directed importer edges; it is explicitly marked `notARetweetCascade=true`. The edge container must not be read as observed retweets, replies, or follower ties. Hydration of the reported 814,924 tweet identifiers did not occur: the accessible OSF identifiers were stored in lossy scientific notation, the Mendeley identifier dump was not retrieved, and no Twitter/X bearer token was available. The fallback therefore cannot reproduce user-level diffusion or temporal ordering. Debnath et al. did not report the present 0–5 misinformation index or MPR, so there is no empirical Twitter MPR against which simulated MPR can be matched.

Mitigation. The graph is labelled as a structural hashtag fallback, tweet IDs are not invented, and simulated auditor scores are never described as observed Twitter misinformation. Future work requires a lawful, lossless identifier source and approved API access, followed by an explicit separation of hashtag co-occurrence, retweet, reply, mention, and follower networks. Any empirical content outcome would also require a separately validated annotation protocol rather than retrofitting simulated MPR.

Distributional validation with $n_{\text{real}} = 1$. The KS and Jensen–Shannon comparisons use one empirical graph against simulated cascades. With $n_{\text{real}} = 1$, a two-sample KS result has little power and does not estimate natural variation among empirical firestorms. The failed DTFS score ($0.2754 < 0.70$) is evidence against the configured validation threshold, not a near-success to be rescued by individual metric matches. Breadth similarity alone does not establish a digital twin.

Mitigation. These quantities are presented as diagnostics and the failed threshold is retained. Future work should collect multiple comparable empirical cascades, define metrics and tolerances before looking at simulation outputs, use uncertainty-aware posterior predictive checks, and validate structure, timing, content, and intervention response separately.

7.0.3 Reproducibility and resource constraints

The repository retains configs, model names, seeds, raw outputs, parser code, derived tables, hatch rules, and the official harvest timestamp. These artefacts support computational traceability: a reader can follow a reported number to a cell and inspect the transformation. They do not guarantee exact reproduction because the external model service, model weights, API defaults, and provider-side moderation can change. Re-running the full campaign also incurs substantial monetary, energy, and time costs, and indiscriminate reruns could overwrite or silently diverge from the recorded harvest.

Mitigation. Phase 2 outputs are isolated from Phase 1, completed runs are preserved, secrets are excluded, and analysis can be regenerated without new LLM calls. A reproducibility release should add immutable checksums, software and package lockfiles, model snapshot identifiers where available, machine-readable provenance, exact cost/token totals, and a small deterministic fixture for testing parsers without paid calls. Replication plans should budget API calls in advance and prioritise statistically informative replicates over a larger but unreplicated grid.

7.0.4 Privacy, research ethics, and dual use

API privacy and cost. Sending content to hosted APIs transfers data to a third party and may create retention, jurisdiction, and terms-of-service obligations. Although this campaign uses curated articles and synthetic personas, future hydration could introduce usernames, tweet text, profile fields, or deleted content. Public availability does not remove contextual privacy or justify redistributing personal data. API access is also financially unequal and can make nominally reproducible research inaccessible to groups without equivalent budgets.

Mitigation. The failed hydration workflow discarded sampled tweet text and user fields after aggregate hashtag extraction, did not fabricate IDs, and did not expose credentials. Any future empirical collection should undergo institutional ethics review, follow platform terms and data-minimisation principles, store identifiers separately with access controls, honour deletion where feasible, publish aggregates instead of raw personal data, document retention, and disclose expected and realised API costs.

Dual-use risk. A simulator that identifies combinations of topology, identity prompts, and message framing associated with high propagation or distortion could support defensive stress testing, but it could also be used to optimise propaganda, harassment, astroturfing, or evasion of moderation. Detailed persona prompts and high-performing cascade recipes can lower the cost of such misuse.

Mitigation. The thesis should report scientific findings without packaging operational targeting instructions, avoid claims about vulnerable real-world groups, and release sensitive artefacts proportionately. Future work should include a formal dual-use review, threat modelling, access tiers for potentially actionable configurations, abuse monitoring, and evaluation of defensive applications such as fact-check timing without deploying content to live platforms.

No human evaluation. Human-evaluation templates exist in run directories, but no completed, ethics-approved human study was conducted. There is therefore no evidence that people find the rewrites realistic, that experts agree with the auditor, or that the personas faithfully represent lived discourse. The absence of human evaluation is a validity limitation and an ethical safeguard: incomplete templates must not be presented as participant data.

Mitigation and future work. A follow-up study should obtain ethics approval before recruitment, use informed consent and fair compensation, minimise exposure to harmful misinformation, provide withdrawal and debriefing procedures, and recruit both subject-matter experts and lay raters for distinct questions. The protocol should preregister realism and factuality rubrics, blind raters to condition, measure inter-rater agreement, and compare human, independent-model, and current-auditor scores. Human review should complement, not merely rubber-stamp, automated scoring.

7.0.5 Priority future work

The most consequential next step is not a larger unreplicated grid. It is a smaller, preregistered validation study with independent seeds, matched compositions, longer horizons, cross-model generation and judging, and blinded human annotation. In parallel, empirical comparison should remain structural until multiple lawfully obtained, timestamped cascades and a validated content measure are available. Only after those conditions are met would it be reasonable to test claims about generalisable topology effects, real-world firestorm timing, or predictive correspondence with social-media misinformation.

8 Conclusion

This thesis examined audited semantic propagation in an LLM-agent society. It connected a measurement instrument to persona composition, topology, a separate complex network, and Pfeffer’s firestorm account. The evidence supports a bounded mechanism study. It does not support a model of real Twitter users.

8.1 Answers to the research questions

RQ1: measurement. The continuous and dual-discrete auditors produced different levels and weakly aligned topology rankings. Dual-discrete MI exceeded continuous MI on all eight topology aggregates in both identity arms. The mean topology gaps were 0.8201 for H and 1.0450 for He. These values show instrument dependence. They do not show that one auditor measured the true amount of misinformation. Continuous and discrete MPR must therefore remain separate.

RQ2: persona composition. Explicit composition described the results more consistently than the H/He label. Homogeneous conspiracy-family means were 2.2194 continuous and 3.1691 dual-discrete. Science and environment means were 0.1118 and 0.8687. The conspiracy-heavy `mix_00` also exceeded conspiracy-free `mix_02` under both instruments. These contrasts apply to the executed prompts and model. They do not establish a general effect of human diversity.

RQ3: topology and tipping. Topology conditioned the composition contrast, but no universal topology ranking emerged. Continuous He–H differences changed sign across the eight graphs. Dual-discrete differences were positive on all eight. Dead-cell rates also varied by topology, and live-only means describe surviving cascades. The observed k^* values are eight-tick, instrument-specific outcomes. They are not empirical ignition times.

RQ4: complex-network generalisation. The 63-node D-net preserved an empirically motivated hashtag structure and supported a controlled composition comparison. Conspiracy-only H exceeded the mixed assignment for both articles under both instruments. Structural correspondence remained partial. Breadth met the selected tolerance, whereas depth and structural virality did not. The Digital Twin Fidelity Score was 0.2754 and validation failed. D-net is therefore a separate boundary test, not a validated retweet cascade or a source of empirical Twitter MPR.

8.2 Contributions and boundary

The thesis contributes a transparent theory-to-design ledger, a 1,728-cell climate factorial, an explicit measurement-sensitivity analysis, a composition-conditioned interpretation, and an honest 63-node boundary test. The canonical harvest contains 288 configurations, 1,728 cells, and 292 hatched-dead cells. It uses `gpt-4o-mini`, eight hops, and one run per configuration. The C5 “combined” result is a table union. It is not a physical supergraph.

Pfeffer, Zorbach, and Carley identified seven Outlook factors (Pfeffer, Zorbach, and Carley, 2014). The six-knob vocabulary in this thesis is a computational remapping. Cross-media dynamics remained held. Empirical timing was unavailable. The engine’s ternary actions did not implement the original binary-choice factor. These omissions prevent a seven-factor confirmation.

8.3 Outlook

The next study should prioritise validation over scale. Independent graph and model seeds should support uncertainty estimates. Matched persona compositions should isolate diversity from prevalence. Independent human coders should assess factuality and realism under an approved protocol. Longer horizons should test censoring at tick eight. Lawfully obtained timestamped interaction data would permit a genuine temporal and cascade comparison. Until those conditions are met, this testbed is best used to probe mechanisms and expose measurement assumptions.

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Reproducibility materials

This appendix records the executable specification of the Phase 2 campaign. Paths are relative to the repository root. The canonical machine-readable design is `thesisExperiment/configs/grid_phase2.json`; where this appendix and a generated artefact disagree, that file and the raw run metadata take precedence. The campaign used one graph seed (`graphRandomSeed: 42`) and one replicate, and therefore is not a replicated inferential experiment. Continuous and dual-discrete misinformation indices are separate instruments and must not be averaged.

1 Configuration matrix

Table 1: Phase 2 factorial configuration matrix. A cell is one configuration–article pair. D-net and probes are outside the 1,728-cell factorial grid.

Slice	Instrument	Identity arm	Topologies	Identity settings	Configs	Article cells
T2c H	continuous	H	8	12 per-sonas	$8 \times 12 = 96$	$96 \times 6 = 576$
T2d H	dual; discrete head-line	H	8	12 per-sonas	$8 \times 12 = 96$	$96 \times 6 = 576$
T2c He	continuous	He	8	6 mixes	$8 \times 6 = 48$	$48 \times 6 = 288$
T2d He	dual; discrete head-line	He	8	6 mixes	$8 \times 6 = 48$	$48 \times 6 = 288$
T2 total	two instruments	H and He	8	18 settings	288	1,728
D-net	continuous or dual	H or He	custom	2 assignments	4	8
Probes	field checks	selected	custom or chain	selected	5 configs	5 parsed rows

All T2 configurations use `gpt-4o-mini` for both the propagating agent and auditor, eight nodes, eight ticks, a maximum of eight hops, one drip seed (`node_0`), an inbox cap of four, an always-active schedule, relation evolution with `trustDelta=0.05`, and six seed articles. Graph configurations use action weights 0.35/0.50/0.15 for forward/reinterpret/drop and trust

threshold 0.15. The linear chain is the deliberately rewrite-heavy exception: 0.10/0.85/0.05 and threshold 0.08. Output is isolated under `thesisExperiment/runs_phase2/`; Phase 1 `runs/`, `results/tables/`, and `configs/full/` are not inputs to the Phase 2 headlines.

Configuration names have the grammar shown in Listing 1. The parser matches the longest known topology token before interpreting the remainder as a persona or mix.

Listing 1: Phase 2 configuration-name grammar.

```
T2{c|d}_{H|He}_{topology}_{persona-id|mix-id}

c  -> miScoringMode = "continuous"
d  -> miScoringMode = "dual" (headline is discrete)
H  -> one homogeneous persona
He -> one heterogeneous eight-seat mix

Examples:
T2c_H_linear_chain_climate_scientist
T2d_He_polarized_mix_00
```

2 Network topology definitions

Table 2: The eight directed, eight-node T2 topology generators.

Identifier	Parameters	Construction
<code>linear_chain</code>	edge trust 0.7	Directed edges $v_i \rightarrow v_{i+1}$ for $i = 0, \dots, 6$.
<code>ring</code>	edge trust 0.7	Linear chain plus $v_7 \rightarrow v_0$.
<code>random_er</code>	$p = 0.42$; minimum seed out-degree 2	Each ordered non-self pair is sampled; a guard adds outgoing seed edges if required.
<code>small_world</code>	$k = 4, \beta = 0.1$	Directed ring lattice to two neighbours on each side; forward-side edges are rewired with probability β .
<code>scale_free</code>	Barabási–Albert $m = 2$	A fully connected $m+1$ seed is extended by preferential attachment; selected old–new pairs receive edges in both directions.
<code>echo_chamber</code>	2 chambers; $p_{\text{in}} = 0.75$, $p_{\text{out}} = 0.08$; trust 0.85/0.15; seed out-degree 2	Alternating chamber membership; dense high-trust within-chamber and sparse low-trust cross-chamber directed edges.
<code>polarized</code>	$p_{\text{in}} = 0.75$, $p_{\text{out}} = 0.05$; trust 0.88/0.10; seed out-degree 2	Two tightly connected halves with sparse, low-trust cross-cluster edges.
<code>hierarchical</code>	branching factor 3; down/up trust 0.85/0.30	Rooted three-ary tree with parent–child and weaker child–parent edges.

Homophily-adjusted generators calculate trust from the Jaccard overlap of persona tags and clamp it to $[0.05, 0.95]$. The stored `graph_topology.json`, rather than the generator label alone, is the authoritative realised graph. D-net is not a ninth T2 generator: it is a 63-node, 228-directed-edge custom graph copied from the reconstructed hashtag co-occurrence object.

3 Persona and article inventory

3.1 Personas

Table 3: Merged Phase 2 persona inventory. “H” marks membership in the 12-person homogeneous grid; the other five personas are available to heterogeneous mixes or graph cluster pools.

Persona identifier	Family/type	H grid	Operational role
<code>conspiracy_believer</code>	conspiracy	yes	high-centrality chemtrails account
<code>conspiracy_haarp_weather</code>	conspiracy	yes	HAARP and weather-warfare frame
<code>conspiracy_depopulation</code>	conspiracy	yes	depopulation and Gates-target frame
<code>conspiracy_climate_iggyback</code>	conspiracy	yes	climate-action hashtag hijack
<code>climate_action_advocate</code>	climate action	yes	mitigation, justice, accountability
<code>climate_justice_youth</code>	climate action	yes	youth and intergenerational justice
<code>mitigation_first_policy</code>	climate action	yes	policy and SRM-governance analysis
<code>environmental_concern</code>	environment	yes	general ecological-risk frame
<code>ozone_stratosphere_specialist</code>	environment	yes	atmospheric chemistry and ozone
<code>biodiversity_food_security</code>	environment	yes	ecosystems, rainfall, and crops
<code>climate_scientist</code>	expert	yes	assessed-science correction
<code>science_journalist</code>	expert/media	yes	sourced factual summary
<code>conspiracy_peripheral_skywatcher</code>	conspiracy	no	low-centrality first-person sky report
<code>climate_action_sweden_scopex</code>	climate action	no	Swedish SCoPEX consent protest
<code>platform_moderator_toxicity</code>	institution	no	procedural harm reduction

Table 3 continued

Persona identifier	Family/type	H grid	Operational role
caregiver_air_quality	env./health	no	household health and air quality
hashtag_attention_communicator	communication	no	high-visibility hashtag amplification

These are theory-faithful reductions of discourse roles, not HDBSCAN centroids estimated from the unavailable 814,924 tweets. The full prompt text and metadata are in `thesisExperiment/personas/merged.json`; compact per-condition files are in `thesisExperiment/personas/phase2/`. The six heterogeneous mixes, in node-seat order, are:

Listing 2: Heterogeneous eight-seat persona mixes.

```

mix_00 = [conspiracy_believer, conspiracy_haarp_weather,
          conspiracy_depopulation, conspiracy_climate_piggyback,
          climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern]
mix_01 = [conspiracy_depopulation, conspiracy_climate_piggyback,
          climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern,
          ozone_stratosphere_specialist, biodiversity_food_security]
mix_02 = [climate_action_advocate, climate_justice_youth,
          mitigation_first_policy, environmental_concern,
          ozone_stratosphere_specialist, biodiversity_food_security,
          climate_scientist, science_journalist]
mix_03 = [mitigation_first_policy, platform_moderator_toxicity,
          ozone_stratosphere_specialist, climate_action_sweden_scopex,
          biodiversity_food_security, climate_scientist,
          conspiracy_believer, climate_justice_youth]
mix_04 = [climate_action_advocate, conspiracy_peripheral_skywatcher,
          science_journalist, biodiversity_food_security,
          climate_action_sweden_scopex, platform_moderator_toxicity,
          ozone_stratosphere_specialist, conspiracy_depopulation]
mix_05 = [climate_scientist, climate_action_advocate,
          conspiracy_climate_piggyback, biodiversity_food_security,
          platform_moderator_toxicity, science_journalist,
          ozone_stratosphere_specialist, conspiracy_peripheral_skywatcher]

```

3.2 Articles

Table 4: Merged article inventory. Every article contains five yes/no ground-truth questions. “Core” marks the six articles used in every T2 configuration.

Article identifier	Domain	Use	Short topic
scopex_2017	geoengineering	core, D-net	SCoPEX measurement proposal

Table 4 continued

Article identifier	Domain	Use	Short topic
chemtrails_gates_2018_2021	misinformation	core, D-net	chemtrails, SCoPEX, and Gates
sai_geoengineering	geoengineering	core	SAI proposal versus deployment
paris_agreement	policy	core	Paris treaty and NDCs
climate_consensus	science	core	anthropogenic-warming consensus
polar_bears	science	core	sea ice and hunting habitat
spice_uk_cancelled	geoengineering	library	cancelled SPICE field test
glaciers_retreat	science	library	global glacier retreat
co2_fertilization	misinformation	library	plant growth versus climate harm
sea_level_rise	science	library	thermal expansion and land ice
net_zero	policy	library	emissions-removals balance
attribution_extremes	science	library	probabilistic event attribution
paris_agreement_2015	policy	library	expanded Paris explainer
consensus_97_percent	science	library	conditional 97% consensus
glacier_retreat	science	library	WGMS glacier mass balance
polar_bears_sea_ice	science	library	IUCN and ice-season evidence
co2_plant_food	science	library	fertilisation and ocean acidification
sai_solar_geoengineeering	geoengineering	library	SAI risks and termination shock
net_zero_2050	policy	library	carbon budgets and mid-century pathways
extreme_weather_attribution	science	library	likelihood and intensity
climatefever_polar_bears	science	library	ClimateFEVER evidence labels
climatefever_co2_claims	science	library	ocean-degassing and greenhouse claims
volcanic_vs_human_co2	science	library	volcanic versus anthropogenic emissions
great_barrier_reef_bleaching	science	library	marine heat and bleaching

The source text, exact questions, expected Boolean answers, and source URLs are stored

in `thesisExperiment/articles/merged.json`. The inventories above identify experimental inputs; they do not reproduce the full stimuli.

4 Prompt templates and IFD fields

Each reinterpretation uses the selected persona’s `systemPrompt` and the fixed user template below. Forward actions make no rewrite call. Agent and auditor outputs are model outputs, not human ratings.

Listing 3: Rewrite user prompt from `src/SimulationNode.js`.

```
Here is the information you received. Rewrite it through your
perspective in 100-200 words. Return only the rewritten text,
no preamble:
```

```
<incoming content>
```

Listing 4: Discrete and continuous auditor templates from `src/Auditor.js`.

DISCRETE SYSTEM

```
You are a fact-checker. For each question, the expected correct
answer is provided.
```

```
Evaluate the given text and return a score for each question:
```

```
1 = text correctly states the expected answer
```

```
0 = text does not contain enough information to address the question
```

```
-1 = text contradicts or distorts the expected answer
```

```
Return only a JSON object: {"answers": [1, 0, -1, ...]}
```

CONTINUOUS SYSTEM

```
You are a fact-checker. For each question, the expected correct
answer is provided.
```

```
Rate how accurately the given text addresses each question on a
scale from 0.0 to 1.0:
```

```
1.0 = text fully and correctly states the expected answer
```

```
0.5 = text does not address this question (missing or neutral)
```

```
0.0 = text directly contradicts or inverts the expected answer
```

```
Intermediate values represent partial accuracy.
```

```
Return only a JSON object: {"scores": [1.0, 0.5, 0.8, ...]}
```

USER, BOTH MODES

```
Text:
```

```
<text>
```

```
Questions (with expected answers):
```

```
1. <question> (expected: Yes|No)
```

```
...
```

For $m = 5$ questions, discrete scores $s_i \in \{-1, 0, 1\}$ produce

$$MI_d = n_{\text{missing}} + n_{\text{incorrect}}, \quad CR = n_{\text{correct}}/m, \quad MR = n_{\text{missing}}/m, \quad IR = n_{\text{incorrect}}/m.$$

Continuous scores $q_i \in [0, 1]$ produce

$$\text{CR} = \frac{1}{m} \sum_i q_i, \quad \text{MI}_c = m(1 - \text{CR}).$$

For descriptive CR/MR/IR counts in continuous mode, $q_i \geq 0.67$ is correct, $q_i \leq 0.33$ is incorrect, and intermediate values are missing. In dual mode two calls are made in parallel. The top-level IFD and `misinfoIndex` mirror the discrete result; the sidecar stores both results, absolute gap $|\text{MI}_d - \text{MI}_c|$, and Pearson agreement after mapping $-1, 0, 1$ to $0, 0.5, 1$.

Table 5: Fields in an event’s `ifd` object.

Field	Type/range	Definition
<code>mi</code>	$[0, 5]$	Discrete count or continuous $m(1 - \text{CR})$.
<code>cr, mr, ir</code>	$[0, 1]$	Correct, missing, and incorrect rates.
<code>cms</code>	non-negative	Discrete $\text{IR}/(\text{CR} + 10^{-9})$; continuous $(1 - \text{CR})/(\text{CR} + 10^{-9})$.
<code>ie</code>	non-negative	Entropy-like score over the mode’s rate terms.
<code>scores</code>	array length 5	Per-question discrete or continuous scores.
<code>mode</code>	string	<code>discrete</code> , <code>continuous</code> , or <code>dual</code> .
<code>dual.discrete</code>	object	Complete discrete IFD object.
<code>dual.continuous</code>	object	Complete continuous IFD object.
<code>dual.gap</code>	float	Absolute MI difference; a diagnostic, not a third MPR.
<code>dual.agreement</code>	$[-1, 1]$ or null	Pearson correlation of aligned question scores.

On auditor JSON parse failure the current engine defaults discrete questions to 1 and continuous questions to 1.0, emitting a warning. This is a consequential implementation choice and is retained here for reproducibility.

5 Raw run schema

Listing 5: Raw run directory layout.

```
runs_phase2/<experimentName>_<YYYY-MM-DD_HH-MM-SS>/
  metadata.json
  state.json
  graph_topology.json
  human_eval_template.csv
  results_<articleId>.json
  nodes/node_0.json ... nodes/node_7.json
```

T2 runs contain six `results_*.json` files. D-net contains two and stores 63 `nodes/user_*.json` files; probes contain one result. The parser selects the newest complete directory with `llmUsage.calls>0` for each `experimentName`.

Table 6: Raw JSON schema by file.

Object	Fields and interpretation
metadata.json	experimentId, status, timestamp, full config, llmUsage, and per-article results. config.experimentName is the reliable name when the optional top-level name is null.
llmUsage	calls, promptTokens, completionTokens, totalTokens, errors, model, and estimatedUsd. No credential is stored.
state.json	experimentId, phase, status, propagation progress, audit progress, error, and lastUpdated.
node_*.json	nodeId, personaId, modelId, outgoing relations, node params, inbox, event history, and action stats.
History event	tick, articleId, sourceNodeId, hops, action, contentIn, contentOut, misinfoIndex, ifd, frameAnalysis, reason, chainTrust, provenance, and timestamp.
results_<article>.json	nodeSummaries, metrics, provenanceMetrics, and networkEvolution. Typical metrics include network MI over time, IFD metrics, structural virality, critical mass, and persona IFD; dual runs add dual metrics.
graph_topology.json	realised node records and directed {from,to,trust} edges.

6 Parsed table schema

The canonical wide schema is shared by TH_rows.csv, The_rows.csv, continuous.csv, dual_discrete.csv, dead_cells.csv, and all_rows.csv.

Table 7: Canonical parsed-cell fields, in CSV column order.

Field	Meaning
runDir	Selected timestamped raw directory.
experimentName	Stable configuration identifier.
slice, arm	T2 slice and H/He arm.
topology, rest	Generator/custom label and persona or mix identifier.
miScoringMode	continuous or dual.
articleId	Seed article; together with experiment name forms the cell key.
status, llmCalls	Run completion state and real model-call count.
nEvents, nScored	Collected events and events with non-null MI.
dead	True iff nScored<=1.
meanMI	Mean event misinfoIndex; the cell headline.
meanNodeMPR	Mean of per-node mean MI; shown beside, not merged with, meanMI.

Table 7 continued

Field	Meaning
<code>maxMI</code>	Maximum scored-event MI.
<code>meanDiscreteMI</code>	Mean discrete MI where available.
<code>meanContinuousMI</code>	Mean continuous MI, including dual sidecar where available.
<code>meanDualGap</code>	Mean absolute dual gap; not an MPR.
<code>meanAgreement</code>	Mean dual per-event Pearson agreement.
<code>kStarDiscrete</code>	First tick at which mean discrete MI exceeds 3 and remains above 3.
<code>kStarContinuous</code>	Continuous equivalent of k^* .
<code>kStarCensoredDiscr</code>	True when discrete k^* occurs at the last observed tick.
<code>ete</code>	
<code>hatchStatus,</code> <code>hatchNote</code>	Live/dead rendering status and interpretation warning.

`dual_gap.csv` is narrow. Its fields are `experimentName`, `articleId`, `topology`, `meanDualGap`, `meanAgreement`, `meanDiscreteMI`, and `meanContinuousMI`. Join it to the wide dual table with experiment name and article identifier as the composite key. Pair T2c with T2d with topology, persona or mix, and article identifier as the composite key. Never join by row number. Dead cells remain in the ledger and are excluded from live means rather than imputed as zero.

7 D-net mapping

The empirical object is `thesisExperiment/data/derived/debnath_hashtag_cascade.json`, schema `thesisExperiment.debnath_hashtag_cascade.v1`. It is a hashtag co-occurrence graph, not a retweet or `conversation_id` cascade; its `retweets[]` array is an importer edge container. It has 63 nodes, 228 directed edges, seed `user_chemtrails_hub`, no empirical MPR, and no empirical time series. D-net scores only `scopex_2017` and `chemtrails_gates_2018_2021`.

Table 8: D-net identity mapping counts.

Layer	Conspiracy	Climate action	Environmental	Expert/other
Empirical cluster identity	28	19	14	2
Intended BP family	28	19	13	3
Mixed D-net persona family	28	19	13	3
Homogeneous D-net	63	0	0	0

The mixed assignment exactly matches the reconstructed intended persona ID for 56 of 63 seats. The seven coarsened assignments preserve discourse type:

Table 9: D-net seats whose mixed persona ID is coarser than the reconstruct mapping.

Node	Hashtag	Intended BP	Mixed seat
user_depopulation_amp	#depopulation	conspiracy_depopulat ion	conspiracy_believer
user_climateaction_piggyba ck_amp	#climateaction	conspiracy_climate_p iggyback	conspiracy_believer
user_climateaction_piggyba ck_peri_1	#climateaction	conspiracy_climate_p iggyback	conspiracy_believer
user_biodiversity_hub	#biodiversity	biodiversity_food_se curity	environmental_conc ern
user_biodiversity_peri_1	#biodiversity	biodiversity_food_se curity	environmental_conc ern
user_foodsecurity_amp	#foodsecurity	biodiversity_food_se curity	environmental_conc ern
user_mitigation_amp	#mitigation	mitigation_first_pol icy	climate_action_adv ocate

The complete 63-row node map is retained in `thesisExperiment/analysis_full/PFE_FFER_DEBNATH_VS_SIM.md`; duplicating it here would create a second canonical mapping. Homogeneous D-net deliberately overwrites all 63 seats with `conspiracy_believer`. H and He use the same edge set, so clustering is fixed while label-dependent homophily changes.

8 Reproduction commands

Run the following from the repository root. Parsing and plotting consume existing runs; the D-net and campaign runners can make paid API calls and are shown only for provenance.

Listing 6: Parse, compare, replot, and compile commands.

```
# Rebuild canonical Phase 2 CSVs and parse summaries.
node thesisExperiment/scripts/parse_phase2.js

# Recompute the structural Debnath/D-net comparison only if its
# empirical topology has changed.
node thesisExperiment/scripts/compare_phase2.js

# Replot both analysis layers.
python3 thesisExperiment/analysis_phase2/plot_phase2.py
python3 thesisExperiment/analysis_full/analyze_full.py

# Recompute the Pfeffer/Debnath mapping overlay without new LLM runs.
node thesisExperiment/scripts/pfeffer_debnath_vs_sim.js

# Compile the existing standalone Phase 2 LaTeX chapter.
make -C thesisExperiment/latex_phase2
# Equivalent:
cd thesisExperiment/latex_phase2
pdflatex -interaction=nonstopmode -halt-on-error main
```

```

biber main
pdflatex -interaction=nonstopmode -halt-on-error main
pdflatex -interaction=nonstopmode -halt-on-error main

# Potentially paid execution commands; normally skip because harvest
# is complete.
node thesisExperiment/scripts/run_dnet.js
node thesisExperiment/scripts/master_phase2.js --skip-probe

```

The present file is an input fragment. Once the full thesis master exists, include it with `\input{drafts/15_APPENDICES}` and compile that master with its normal `latexmk` or `pdflatex/biber` sequence. No master file is assumed in this draft.

9 Artifact index

Table 10: Canonical Phase 2 artefact paths.

Path	Purpose
thesisExperiment/configs/grid_phase2.json	machine-readable design and inventories
thesisExperiment/configs/phase2/	288 T2 configs, four D-net configs, index, and probes
thesisExperiment/personas/merged.json	17-person merged library
thesisExperiment/personas/phase2/	exact H and He persona files
thesisExperiment/articles/merged.json	24 articles with questions and expected answers
thesisExperiment/runs_phase2/	raw Phase 2 runs and status records
thesisExperiment/results_phase2/tables/	canonical parsed CSVs
thesisExperiment/results_phase2/summary.json	harvest counts and generated-at stamp
thesisExperiment/results_phase2/PARSE.md	human-readable parse ledger
thesisExperiment/results_phase2/debnath_compare.json	structural comparison
thesisExperiment/analysis_phase2/	first analysis and plots
thesisExperiment/analysis_full/	C1–C5 tables, plots, notes, and key numbers
thesisExperiment/analysis_full/PFEFFER_DEBNATH_VS_SIM.md	full D-net map and observable audit

Table 10 continued

Path	Purpose
<code>thesisExperiment/data/derived/debnath_hashtag_cascade.json</code>	63-node reconstructed custom graph
<code>thesisExperiment/latex_phase2/</code>	standalone Phase 2 results chapter
<code>thesisExperiment/AGENT_HANDOFF.md</code>	detailed schema and harvest handoff
<code>thesisExperiment/CONSOLIDATION_LOG.md</code>	branch provenance, overlap resolutions, and canonical 292-dead-cell decision
<code>thesisExperiment/CONSOLIDATED_OVERLAPS/</code>	retained non-canonical conflict copies

10 Credential handling and consolidation provenance

The OpenAI credential is runtime infrastructure, not a research artefact. It must be supplied through `OPENAI_API_KEY` or an ignored local `.env` file. Never paste a value into a configuration, manuscript, terminal transcript, issue, manifest, or committed file; never print `.env`. The safe readiness check is

```
node thesisExperiment/scripts/check_openai_key.js
```

which reports only presence, length, and candidate locations. Repository ignore rules cover `.env`, `.env.local`, and `.env.*.local`. Raw metadata records usage and model names but no API credential.

The consolidation record is `thesisExperiment/CONSOLIDATION_LOG.md`. It identifies source branches, merge order, retained overlap copies, official artefact locations, and the resolution of the historical 290-versus-292 dead-cell discrepancy. The canonical parse ledger contains 292 dead T2 cells; older 290-cell analysis copies remain under `thesisExperiment/CONSOLIDATED_OVERLAPS/analysis_phase2_hatch290/` for auditability and must not replace the canonical tables.